

When Does Disclosure Help? Information and Coordination in Equity Financing*

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Abstract

Policies such as mandatory financial disclosure for private firms change how much investors know about the firms they fund, and whether that knowledge is public or private. This paper asks when more information about firms raises welfare in the context of equity financing. I develop a model in which investors choose how much capital to contribute to each firm in exchange for a share of its future profits, observing only a private and a public signal of its productivity. How an investor's contribution responds to the contributions of other investors is governed by two opposing forces. Larger contributions by other investors raise the profits that investors share, which pushes an investor to contribute more, but they also dilute its share of those profits, which pushes it to contribute less. The balance between these forces determines whether investors react to one another too much or too little. I show that more public information can lower welfare when investors react too much to what other investors are doing, and that more private information can lower welfare when they react too little. Finally, if a policymaker can decide only whether information is public or private, not how much there is, it is optimal to make all information public or all of it private, and I characterize when each is optimal.

Keywords: disclosure, social value of information, coordination, equity financing

JEL Classification: D62, D82, D83, G32, G38

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1 Introduction

In the United States, private firms are generally not required to publish their financial statements or to have them audited, whereas many private firms in most other countries must file at least some financial information publicly ([Minnis and Shroff, 2017](#)). In a 2017 speech, Securities and Exchange Commission (SEC) Commissioner Kara Stein asked whether enhanced disclosure laws should apply to companies that stay private, or if such companies require rules of their own ([Stein, 2017](#)). In 2022 the SEC began work on a plan that would require private companies to report regularly on their finances and operations ([Kiernan, 2022](#)). Whether such firms should be required to disclose more about themselves is an open policy question.

The case for mandatory disclosure rests on the premise that investors who fund a private firm know too little about it, and that more information is always beneficial. A mandate, however, does more than give investors more information. It makes that information public. In equity financing this matters, because a firm raises capital from many investors at once and their decisions are strategically linked. How much one investor wants to contribute depends on how much it expects the others to contribute, and hence on what it believes they know. A disclosure mandate therefore changes not only how much investors know about a firm, but also how much they know about what other investors know. This paper asks when more information about firms raises welfare in equity financing, and whether it matters if that information is public or private.

I answer this question with a model of equity financing in which a continuum of investors fund each firm without observing its productivity, relying instead on a private and a public signal about it. The firm produces with the capital its investors provide. In exchange, it pays them a fraction of its output, which I call the payout, and each investor's share of the payout depends on how much it contributed. Two parameters describe how investors' payoffs are linked. The first is how substitutable their capital is in production. The second is how strongly an investor's share of the payout rises with its own contribution. Investors coordinate too much when an investor's share rises little with its contribution relative to how substitutable their capital is, and in that case more public information can reduce welfare. When it rises a lot, they coordinate too little, and more private information can reduce welfare.

The economy lasts two periods. A continuum of firms, each with a productivity drawn from a common distribution, raises capital in the first period by issuing equity and produces in the second. A continuum of investors funds them. Investors obtain funds from a risk neutral household and provide capital to the firms, and a government can subsidize that cost, financing the subsidy with a lump-sum tax on the household. Firms hold no resources of their own, so the capital they use is what investors provide.

A firm combines the capital contributions of its investors through a CES aggregator, and produces with decreasing returns to capital. Before productivities and signals are realized, it chooses how much equity to sell, the fraction of its output that goes to investors as a group. How that fraction is divided among the investors is governed by a sharing rule. I restrict attention to sharing rules that satisfy three properties (shares are nonnegative and exhaust the payout, only relative contributions matter, and each investor's share depends on its own contribution through a function common to all investors), which pin down the rule up to a single parameter, the elasticity of an investor's share with respect to its own contribution. I call it the share elasticity.

Investors do not observe productivity. Each investor receives a private signal, which only it observes, and a public signal, which every investor observes. The public signal does two things at once. It informs an investor about the firm, and, because every other investor sees the same realization, it also tells the investor what the others know. It is the channel through which investors' beliefs, and hence their contributions, become correlated. An investor choosing how much to contribute therefore faces two kinds of uncertainty: about the firm's productivity and about how much the other investors will contribute. The contributions of the other investors matter because they affect an investor's payoff in two opposing ways. A larger contribution by the others raises the firm's capital and with it the payout that investors share, which pushes an investor to contribute more. It also dilutes the investor's share of that payout, which pushes it to contribute less.

The balance between these two effects determines the degree of strategic complementarity, how strongly an investor's contribution responds when every other investor scales up its own. The returns to capital govern the first effect and push toward complementarity, while the share elasticity governs the second and pushes toward substitutability. As a result, contributions are strategic complements when the returns to capital exceed the share elasticity, and strategic substitutes otherwise.

To assess efficiency, I compare the equilibrium with the allocation chosen by a planner who determines each investor's contribution directly but cannot pool the information dispersed across investors, so that a contribution may depend only on that investor's own signals. Any difference between the two allocations reflects how the available information is used, not how much of it there is. The equilibrium attains the constrained efficient allocation if and only if two conditions hold. First, an investor's share responds to its contribution exactly as it would if the investor were paid its marginal product, so that investors weight their signals exactly as the planner would. Second, the subsidy is set to offset investors' market power.

I then turn to the main question of the paper, when improving investors' information raises welfare. When the two efficiency conditions hold, more information of either kind raises welfare. This is the counterpart, in equity financing, of the efficient-use-of-information

result of [Angeletos and Pavan \(2007\)](#). The model can be read as a microfoundation of an economy of the kind they study, with a technology, a sharing rule and log-normal productivity in place of an abstract payoff. The two efficiency conditions are restrictive, however. The share elasticity describes how a firm divides its payout among the investors that fund it, whereas the substitutability of their capital is a feature of production. There is no reason for the two to coincide, and when they do not, the welfare effects of public and private information become ambiguous.

Away from that benchmark, more public information raises welfare if and only if its precision relative to that of private information exceeds a threshold. The threshold is positive only when investors coordinate substantially more than is efficient. In that case, if public information is imprecise relative to private information, a more precise public signal leads investors to rely on it even more heavily, amplifies the inefficient comovement of their contributions, and lowers welfare. This result nests that of [Morris and Shin \(2002\)](#). In their beauty contest the planner attaches no value to coordination, and setting the socially optimal degree of coordination to zero here recovers their threshold exactly.

The effect of private information mirrors that of public information. More private information raises welfare if and only if its precision relative to that of public information exceeds a threshold, which is positive only when investors coordinate substantially less than is efficient. In that case, if private information is imprecise relative to public information, a more precise private signal disperses investors' contributions further and moves the allocation away from the one the planner would choose.

Finally, I hold the total amount of information fixed and ask how it should be divided between public and private sources. Mixing the two is never optimal. Except in a knife-edge case in which neither investors nor the planner has any motive to coordinate, and the division does not matter, information should be entirely public or entirely private. It should be entirely public whenever the planner would want investors to coordinate, regardless of how much information there is or how precise the prior is, and entirely private when the planner's preference for investors not to coordinate is strong enough. In the limit of abundant information the rule is simply that information should be public if and only if the planner would want investors to coordinate.

Related Literature. This paper contributes to several strands of the literature.

The first is the literature on the social value of information in economies with dispersed information. [Morris and Shin \(2002\)](#) show that in a beauty contest, in which players want to coordinate but coordination has no social value, more precise public information can lower welfare when private information is sufficiently precise. [Angeletos and Pavan \(2007\)](#) extend the analysis to a class of economies with quadratic payoffs and show that the welfare

effects of information turn on two comparisons. The first is whether the equilibrium is efficient when the fundamental is common knowledge. The second is whether the equilibrium degree of coordination equals the socially optimal one, in which case information is used efficiently. When both hold, more information of either kind raises welfare. When the degree of coordination is inefficient, more information of one kind can lower welfare, and they give sufficient conditions for each kind to be beneficial.

My model can be read as a microfoundation of an economy in their class, in which payoffs follow from a CES aggregator of investors' capital, a sharing rule and log-normal productivity rather than from a quadratic loss. Both degrees of coordination then come from primitives. In my model the equilibrium is efficient under common knowledge when the subsidy offsets investors' market power, and information is used efficiently when an investor's share responds to its contribution exactly as it would if the investor were paid its marginal product. Away from these conditions I characterize the sign of the welfare effect of each signal.

The model also speaks to a question that [Angeletos and Pavan \(2007\)](#) leave open. In their canonical economy with investment complementarities coordination is never excessive, so more public information always raises welfare, and they note that whether models of financial frictions share this property is an open question. In equity financing, I show that coordination can be either too strong or too weak.

[Angeletos, Iovino and La'O \(2016\)](#) study whether welfare improves when firms are better informed about the state of the economy in a microfounded business-cycle model in which dispersed information is the source of both real and nominal rigidity. My analysis draws on theirs in two ways. First, I combine investors' capital through a CES aggregator, as their economy combines differentiated goods. Second, to study the optimal composition of information, I follow them in holding the overall precision of information fixed and asking how it should be divided between private and public sources.

My result on the optimal composition of information relates to papers that ask how public information should be released. [Amador and Weill \(2010\)](#) also obtain a corner solution, since a government should release either all or none of its information, but the corner concerns how much to release and arises because public releases reduce the information that prices aggregate. [Cornand and Heinemann \(2008\)](#) show that a public signal should always be fully precise but sometimes reach only part of the population, which in effect mixes public and private information. In my model the total amount of information is fixed, and mixing the two sources is never optimal. I take the precision of each signal as given, while [Hellwig and Veldkamp \(2009\)](#), [Myatt and Wallace \(2012\)](#) and [Colombo, Femminis and Pavan \(2014\)](#) study how the acquisition of private information responds to coordination motives and to public information.

This paper also relates to the literature on disclosure and real efficiency in financial

markets.¹ The closest paper is [Goldstein and Yang \(2019\)](#). In their model a representative capital provider, which they interpret as banks, equity investors or venture capital firms, decides how much to invest in a firm after observing a public disclosure and the price of the firm's stock, which aggregates the private information of speculators. Disclosure informs the capital provider directly, but it also leads speculators to trade less on their own information about the disclosed factor. When the disclosure concerns what the capital provider most wants to learn and the market aggregates information well, the price becomes so much less informative that better disclosure lowers real efficiency.

Other papers show that disclosure can make prices less informative by crowding out the private acquisition of information ([Gao and Liang, 2013](#)), by attracting noise trading ([Han, Tang and Yang, 2016](#)), or by shifting traders' learning from fundamentals toward the beliefs of other traders ([Banerjee, Davis and Gondhi, 2018](#)), and that a government that intervenes on the basis of prices can reduce what they reveal ([Bond and Goldstein, 2015](#)). Coordination also arises in this literature, but among speculators rather than capital providers. When a capital provider learns from the price, speculators' trades become strategic complements, and they coordinate exactly when coordination is undesirable ([Goldstein, Ozdenoren and Yuan, 2013](#)). Even when disclosure raises real efficiency, it can harm the investors it is meant to protect ([Kurlat and Veldkamp, 2015](#)). I study the same question through a different channel. There is no price to learn from in my model. Instead, the representative capital provider is replaced by a continuum of investors who fund the same firm and whose contributions are strategically linked.

Finally, the paper relates to the literature on information sharing in credit markets and open finance. In the canonical model of [Pagano and Jappelli \(1993\)](#), lenders choose to share information about borrowers through credit bureaus, and sharing raises lending when adverse selection is severe enough that safe borrowers would otherwise leave the market. Open finance extends this logic by letting customers share their own data with the providers they choose. [He, Huang and Zhou \(2023\)](#) show that such sharing promotes competition when it levels the playing field in screening borrowers, but can leave all borrowers worse off when it gives fintech lenders too large an advantage. Empirically, open banking has increased investment in fintechs and enabled new lending relationships ([Babina et al., 2025](#)), and has allowed small businesses to pledge assets such as receivables and inventory as collateral ([Yu, 2024](#)). In Brazil, expanding the information behind credit scores shifted credit toward borrowers deemed more creditworthy and lowered default rates ([Rocha et al., 2024](#)).

¹See [Goldstein and Yang \(2017\)](#) for a review of this literature and [Bond, Edmans and Goldstein \(2012\)](#) for a review of the real effects of financial markets.

Outline. The rest of the paper is organized as follows. Section 2 develops the model. Section 3 defines the degree of strategic complementarity, characterizes the equilibrium and how much equity the firm optimally sells, and derives the conditions under which the equilibrium is constrained efficient. Section 4 characterizes the welfare effects of public and private information and the optimal composition of information. Section 5 concludes. Proofs are in Appendix A.

2 Model

The economy lasts two periods and is inhabited by three types of agents: a continuum of firms, a continuum of investors, and a representative household. Firms differ in their productivity, raise equity from investors to finance capital and, in return, hand over a fraction of their profits. Investors fund each firm without observing its productivity, and choose their contributions based on a private signal, which only they observe, and a public signal, which every investor observes. The household owns the economy's resources, supplies savings to investors, and consumes. A government sets fiscal policy and runs a balanced budget.

2.1. Environment

Time and goods. Time is discrete and there are two periods, $t \in \{1, 2\}$. In each period there is a single good that serves both as the consumption good and, when supplied to firms, as the capital used in production.

Firms. There is a unit mass of firms indexed by $j \in J = [0, 1]$. Firm j is characterized by a productivity z_j , drawn independently across firms from a common distribution. Firms hold no resources of their own. In period 1, firm j raises capital by issuing equity to investors. In period 2 it produces using that capital and distributes its profits to the founder and investors.

Investors. There is a unit mass of investors indexed by $i \in I = [0, 1]$. Investors stand between the household and firms: they obtain funds from the household at a cost of capital R and use them to provide capital to the firms. An investor does not observe a firm's productivity z_j and instead receives a private and a public signal about it.

Household. There is a risk neutral representative household which extracts utility from consumption in both periods. It is endowed with e units of the period-1 good, which it divides between consumption and savings. The household owns the firms and the investors, so in period 2 it collects their profits, net of a lump-sum tax.

Government. A government sets fiscal policy and runs a balanced budget. It subsidizes investors' cost of capital at rate τ , so that an investor effectively pays $(1 - \tau)R$ per unit of funds, and finances the subsidy with the lump-sum tax on the household. The purpose of this instrument is to potentially correct the distortion that investors' market power introduces into the funding of firms.

Timing. The sequence of events is the following.

- (i) At the start of period 1, before productivities and signals are realized, each firm chooses the outside equity stake, the fraction of its output that it will pay to investors as a group.
- (ii) The productivities $\{z_j\}_{j \in J}$ are then realized, and each investor observes a private and a public signal about every firm's productivity.
- (iii) Still in period 1, investors simultaneously choose their capital contributions across firms and the household chooses consumption and savings. The markets for the period-1 good and for capital clear.
- (iv) In period 2, each firm produces and divides its profits among its investors and the founder. The household consumes and pays the lump-sum tax, and the market for the period-2 good clears.

2.2. Firms

Firm j raises capital from the continuum of investors. Let $k_{i,j}$ be the amount of capital that investor i provides to firm j . The firm combines these contributions into effective capital through a CES aggregator:

$$K_j = \left[\int_I k_{i,j}^\rho di \right]^{1/\rho}, \quad \rho \in (0, 1] \quad (1)$$

The parameter ρ sets how substitutable investors' contributions are, with a corresponding elasticity of substitution $1/(1 - \rho)$. When $\rho < 1$ capital from different investors is an imperfect substitute, so effective capital depends on the distribution of contributions across investors, not only on the total raised. When $\rho = 1$ capital contributions are perfect substitutes, money is just money, and only the total funding matters.

In period 2 the firm produces a single good using effective capital, with output

$$z_j K_j^\alpha, \quad \alpha \in (0, 1)$$

where z_j is its productivity and α reflects decreasing returns to capital. The firm issues equity to finance production: investors together receive a fraction $\lambda \in [0, 1]$ of output, the outside equity stake, and the founder keeps the rest, as in [Goldstein and Yang \(2019\)](#). The payout to all investors who invested in firm j is

$$\lambda z_j K_j^\alpha$$

and the founder keeps $(1 - \lambda) z_j K_j^\alpha$.

The firm chooses λ before productivities and signals are realized, so as to maximize the founder's expected payoff,

$$\max_{\lambda \in [0, 1]} \mathbb{E}[(1 - \lambda) z_j K_j(\lambda)^\alpha] \quad (2)$$

Effective capital is written $K_j(\lambda)$ to stress that the stake offered to investors shapes how much they invest, and the firm internalizes this response when choosing λ . Because the investors' response is itself an equilibrium object, I characterize the optimal stake together with the investment equilibrium in [Section 3](#).

2.3. The Profit-Sharing Rule

Firm j owes its investors the payout $\lambda z_j K_j^\alpha$. How this payout is divided among them is governed by a sharing rule. Let $k_{-i,j}$ denote the profile of contributions of all investors other than i to firm j . Investor i receives a share

$$s_{i,j} = s(k_{i,j}, k_{-i,j})$$

and thus collects $s_{i,j} \lambda z_j K_j^\alpha$ of the firm's output. I restrict attention to sharing rules that satisfy three properties.

Assumption 1. The sharing rule satisfies:

- (i) *Feasibility*: $s_{i,j} \geq 0$ for every i , and $\int_I s_{i,j} di = 1$.
- (ii) *Scale invariance*: $s(tk_{i,j}, tk_{-i,j}) = s(k_{i,j}, k_{-i,j})$ for every $t > 0$.
- (iii) *Separability*: there is an increasing and continuous function f such that

$$s_{i,j} = \frac{f(k_{i,j})}{\int_I f(k_{\ell,j}) d\ell}$$

Feasibility says that shares are nonnegative and exhaust the payout. Scale invariance requires that scaling every investor's contribution by a common factor leave the division unchanged, so that only relative contributions matter. Separability requires that an investor's

share depend on its own contribution through a common function f , so investors are ranked by $f(k_{i,j})$ alone.² Together these properties pin down the sharing rule up to a single parameter.

Lemma 1. *Under Assumption 1, the sharing rule takes the power form*

$$s_{i,j} = \frac{k_{i,j}^\gamma}{\int_I k_{\ell,j}^\gamma d\ell} \quad (3)$$

for some $\gamma > 0$.

Proof. See Appendix A.1. □

This is the success function studied by Skaperdas (1996) in the theory of contests. I impose diminishing returns throughout.

Assumption 2. $\gamma \in (0, 1)$.

Assumption 2 ensures that the investor problem is concave and well defined. Because a single investor has measure zero, the denominator in Eq. (3) is unaffected by its own choice, so γ is the elasticity of investor i 's share with respect to its own contribution. I will refer to γ as the share elasticity. When $\gamma = 1$ the rule is proportional and each investor's share equals its share of total funding. When $\gamma \in (0, 1)$ the share is concave in own contribution, so an investor's marginal claim on the payout falls as it invests more. In the limit $\gamma \rightarrow 0$ this concavity becomes extreme and the rule approaches an equal division: since $k_{i,j}^\gamma \rightarrow 1$ for every positive contribution, all investors that fund the firm receive the same share of the payout, however much or little each provides. Only whether an investor participates then matters, not the size of its contribution.³ Together with the returns to capital α , the parameter γ determines investors' incentive to coordinate. I make this precise in Section 3.

2.4. Information

Firm productivity is unknown to investors at the time they invest. It is log-normally distributed and independent across firms,

$$\ln z_j \sim \mathcal{N}(\mu_z, 1/\kappa_z)$$

²Separability implies independence of irrelevant alternatives. For any two investors ℓ and m funding firm j , the ratio of their shares $s_{\ell,j}/s_{m,j} = f(k_{\ell,j})/f(k_{m,j})$ depends only on their own contributions, and not on how much any other investor provides.

³The sharing rule also admits a behavioral microfoundation. If the founder allocates equity through a discrete-choice problem with idiosyncratic taste shocks over investors, the same power form arises, with γ measuring the strength of the founder's valuation of contributions relative to the shocks. When the shocks dominate, $\gamma \rightarrow 0$ and the allocation becomes random, which is the equal division here. See Appendix B.1.

where μ_z is the mean of log productivity and κ_z is the precision of this prior, that is, the reciprocal of its variance.

Each investor i observes two signals about firm j 's productivity. The first is a private signal

$$x_{i,j} = \ln z_j + v_{i,j}, \quad v_{i,j} \stackrel{iid}{\sim} \mathcal{N}(0, 1/\kappa_x)$$

where the noise $v_{i,j}$ has precision κ_x and is independent across investors and firms. The second is a public signal

$$y_j = \ln z_j + \varepsilon_j, \quad \varepsilon_j \stackrel{iid}{\sim} \mathcal{N}(0, 1/\kappa_y)$$

where the noise ε_j has precision κ_y and is independent across firms. All noise terms are mutually independent and independent of productivity.

The private signal is idiosyncratic, since each investor draws its own noise, whereas the public signal y_j is observed by every investor considering firm j . The public signal is common knowledge, so observing it does two things at once: it informs an investor about the firm's productivity, and, since every other investor sees the same realization, it also reveals information about what other investors know about the firm. The public signal is therefore the channel through which investors' beliefs, and hence their investments, become correlated. The private signal works in the opposite direction: being idiosyncratic, it pulls investors' assessments apart. This contrast, between public information as a coordinating device and private information as a dispersing one, is what will make their welfare effects distinct.

In what follows I refer to investor i 's information about firm j as its information set,

$$\omega_{i,j} = (x_{i,j}, y_j) \in \Omega$$

where $\Omega = \mathbb{R}^2$ is the space of possible information sets.

2.5. Investors

Consider investor i deciding how much capital to provide to firm j . Having observed $\omega_{i,j}$, the investor does not know z_j and chooses a contribution $k \geq 0$ to maximize its expected profit. The investor's problem is:

$$\max_{k \geq 0} \mathbb{E} \left[\underbrace{\frac{k^\gamma}{\int_I k_{\ell,j}^\gamma d\ell}}_{\text{sharing rule } s_{i,j}} \cdot \underbrace{\lambda z_j K_j^\alpha}_{\text{payout to investors}} - (1 - \tau) R k \mid \omega_{i,j} \right] \quad (4)$$

The first term in the expectation is investor i 's share of the firm's payout. It is the product of the sharing rule $s_{i,j}$, which depends on the investor's own contribution and on those of all

others, and the total payout to investors $\lambda z_j K_j^\alpha$, which depends on the firm's productivity and on the effective capital K_j that all investors raise. The second term is the cost of the funds to the investor, where R is the cost of capital and τ the subsidy/tax set by the government.

Because each investor has measure zero, it takes the denominator of the sharing rule $\int_I k_{\ell,j}^\gamma d\ell$ and effective capital K_j as given. The first-order condition of Eq. (4) then delivers the best response:

$$k^{1-\gamma} = \frac{\gamma \lambda}{(1-\tau)R} \mathbb{E} \left[\frac{z_j K_j^\alpha}{\int_I k_{\ell,j}^\gamma d\ell} \middle| \omega_{i,j} \right] \quad (5)$$

In funding firm j , investors play a Bayesian game: each observes its own information set, chooses a capital contribution, and earns a payoff that depends on the contributions of all others through the payout and its division. An investor thus faces two sources of uncertainty. The first is *fundamental*: it does not observe the firm's productivity z_j . The second is *strategic*: it does not know how much the other investors will contribute, because each of them observes a different private signal and hence forms a different view of the firm.

How much the others invest matters because it affects investor i through two opposing channels. When other investors contribute more, effective capital K_j rises, and with it the firm's output and the payout $\lambda z_j K_j^\alpha$ that investors share. This raises the return to investing and pushes investor i to contribute more. At the same time, a larger contribution by others dilutes investor i 's share $s_{i,j}$ of that payout, which pushes it to contribute less. Which channel dominates determines the strategic interaction. If the first dominates, investor i wants to invest more the more it expects others to invest, and contributions are strategic complements. If the dilution channel dominates, it wants to invest less, and contributions are strategic substitutes. I make this precise in Section 3.

The right-hand side of Eq. (5) depends on investor i only through its information set. Effective capital K_j and the denominator of the sharing rule $\int_I k_{\ell,j}^\gamma d\ell$ are common to all investors funding firm j , and because each investor has measure zero its own choice does not affect them. As the private signals are identically distributed, the conditional expectation is the same function of the information set for every investor. Two investors with the same information set therefore make the same contribution. The equilibrium is thus symmetric without any restriction: the contribution is a function $k : \Omega \rightarrow \mathbb{R}_+$ of the information set alone. Substituting $k(\omega_{i,j})$ into the best response, the equilibrium strategy solves the following fixed-point condition, for every $\omega_{i,j} \in \Omega$,

$$k(\omega_{i,j})^{1-\gamma} = \frac{\gamma \lambda}{(1-\tau)R} \mathbb{E} \left[\frac{z_j K_j^\alpha}{\int_I k(\omega_{\ell,j})^\gamma d\ell} \middle| \omega_{i,j} \right] \quad (6)$$

where $K_j = \left[\int_I k(\omega_{\ell,j})^\rho d\ell \right]^{1/\rho}$.

2.6. Household

The representative household is risk neutral⁴ and values consumption in the two periods, discounting period-2 consumption by $\beta \in (0, 1)$. It is endowed with e units of the period-1 good, which it divides between consumption and savings. It saves by renting capital $\bar{K}$ to investors at rate R per unit, and capital depreciates at rate δ . The household owns the firms and the investors, so in period 2 it also collects their profits, and it pays the lump-sum tax T .

The household chooses consumption c_1, c_2 and savings $\bar{K}$ to solve⁵

$$\max_{c_1, c_2, \bar{K}} c_1 + \beta c_2 \quad (7)$$

subject to

$$\begin{aligned} c_1 + \bar{K} &\leq e, \\ c_2 + T &\leq (1 - \delta + R)\bar{K} + \Pi^F + \Pi^I \end{aligned}$$

where Π^F collects the founders' retained profits, the fraction $1 - \lambda$ of each firm's output, and Π^I collects the investors' profits, the share of firm payouts each investor receives net of the cost of the funds it provides.

Because the household is risk neutral, its savings decision pins down the cost of capital. An interior optimum⁶ requires $\beta(1 - \delta + R) = 1$, so

$$R = \frac{1}{\beta} - (1 - \delta) \quad (8)$$

2.7. Equilibrium

Having described each agent's problem, I now define the equilibrium of the economy.

Definition 1. An equilibrium consists of an investor strategy $k : \Omega \rightarrow \mathbb{R}_+$, a household allocation $(c_1, c_2, \bar{K})$, an outside equity stake λ , a cost of capital R , and a fiscal policy (τ, T) such that

⁴I relax the assumption of risk neutrality in Appendix D.

⁵Although each firm's productivity is idiosyncratic, the continuum of firms and investors makes all aggregates deterministic. Therefore, there is no aggregate uncertainty, so the household's period-2 resources are known and its problem is written without expectations.

⁶This is the only cost of capital consistent with equilibrium. Because the household is risk neutral, its supply of savings is infinitely elastic at this value of R . At any higher R , saving is worth more than consuming in period 1, so the household would save its entire endowment, more than investors demand at that R , and the excess supply drives R down. At any lower R , the household saves nothing, while investors' demand for capital remains strictly positive, and the excess demand drives R up. Only at the interior value is the household willing to supply exactly what investors demand.

- (i) $k(\cdot)$ is a Bayes–Nash equilibrium of the investors’ game, that is, it solves Eq. (6);
- (ii) $(c_1, c_2, \bar{K})$ solve the household’s problem in Eq. (7);
- (iii) λ solves the firm’s problem in Eq. (2);
- (iv) markets clear and the government balances its budget:

$$\begin{aligned}\bar{K} &= \int_I \int_J k(\omega_{i,j}) dj di \\ c_1 &= e - \bar{K} \\ c_2 &= \int_J z_j K_j^\alpha dj + (1 - \delta)\bar{K} \\ T &= \tau R \bar{K}\end{aligned}$$

Condition (iv) consists of the market clearing for capital, under which household savings equal the total funding that investors provide, the resource constraints of the two periods, and the government’s balanced budget, under which the lump-sum tax finances the subsidy to investors.

3 Equilibrium and Constrained Efficiency

I now solve the model in closed form. I first define the degree of strategic complementarity among investors, then characterize their equilibrium investment strategy and the firm’s optimal outside equity stake. I then turn to efficiency and solve the problem of a planner who allocates capital but, like investors, cannot pool the information dispersed across them, and I derive the conditions under which the decentralized equilibrium is constrained efficient.

3.1 Strategic Complementarity and the Equilibrium Strategy for Investors

An investor’s incentive to contribute depends on how much the others contribute. I measure this dependence by the elasticity of an investor’s best response to a uniform scaling of all other investors’ contributions, holding beliefs fixed.

Definition 2 (Degree of strategic complementarity). Fix investor i ’s beliefs, and suppose every other investor scales its contribution by a common factor $t > 0$. The *degree of strategic complementarity* θ is the elasticity of i ’s best response to this scaling.

In the present model this elasticity is constant and admits a closed form. Effective capital is homogeneous of degree one, so $K_j \rightarrow tK_j$, while the denominator of the sharing rule scales as

$\int_I k^\gamma d\ell \rightarrow t^\gamma \int_I k^\gamma d\ell$. The right-hand side of investor i 's optimality condition Eq. (6) therefore scales by $t^{\alpha-\gamma}$. Because i 's own contribution enters the numerator of its share through k^γ , we have $k^{1-\gamma} \propto t^{\alpha-\gamma}$, and hence

$$\theta = \frac{\alpha - \gamma}{1 - \gamma} \quad (9)$$

The degree of strategic complementarity θ reflects the balance between two opposing forces. A larger aggregate contribution raises effective capital and hence the firm's payout, which raises the return to investing. This force, governed by the returns to capital α , pushes toward complementarity. At the same time, a larger contribution by others dilutes an investor's share of the payout. This force, governed by the share elasticity γ , pushes toward substitutability. Which force dominates depends on the sign of $\alpha - \gamma$. When $\alpha > \gamma$, contributions are strategic complements, $\theta > 0$, and an investor optimally invests more when others do. When $\alpha < \gamma$, they are strategic substitutes, $\theta < 0$, and an investor optimally invests less when others invest more. In the knife-edge case $\alpha = \gamma$, there is no strategic interaction. In all cases $\theta < 1$, since $\alpha < 1$ and $\gamma < 1$.

When contributions are complements, investors wish to align their choices, and so they value information that others also observe. This coordination motive is what makes the composition of public and private information matter for welfare, as in the social-value-of-information literature (Morris and Shin, 2002; Angeletos and Pavan, 2007; Angeletos, Iovino and La'O, 2016).

I can now characterize the equilibrium strategy.

Proposition 1. *There is a unique equilibrium in which every investor uses a strategy that is log-linear in the private and public signals:*

$$\ln k(\omega_{i,j}) = \phi_0 + \phi_x x_{i,j} + \phi_y y_j \quad (10)$$

with loadings

$$\phi_x = \frac{1}{1 - \alpha} \cdot \frac{(1 - \theta)\kappa_x}{\kappa_z + \kappa_y + (1 - \theta)\kappa_x}, \quad \phi_y = \frac{1}{1 - \alpha} \cdot \frac{\kappa_y}{\kappa_z + \kappa_y + (1 - \theta)\kappa_x} \quad (11)$$

where θ is given by Eq. (9).

Proof. See Appendix A.2. □

The intercept ϕ_0 collects the terms that do not vary with the signals and its closed form is in Appendix A.2. It is increasing in the outside equity stake λ , so a more generous stake raises investment uniformly across information sets. A larger stake therefore attracts more capital but leaves the founder a smaller fraction of output, and the founder balances the two when choosing λ .

The loadings in Eq. (11) show how investors weight their two signals. Each weight rises with the precision of the corresponding signal, and decreases with the precision of the other signal. The degree of strategic complementarity tilts the balance between them: as θ rises, the term $(1 - \theta)\kappa_x$ falls, so investors place less weight on their private signal and relatively more on the public signal. When contributions are stronger complements, each investor cares more about aligning with others, and the public signal, being common knowledge, is the better coordination device. This is the overreaction to public information familiar from [Morris and Shin \(2002\)](#). When investors are coordinating more than is efficient, as I make precise in Section 3.2, θ is too large, so they overweight the public signal and underweight the private signal, which is the source of inefficiency. An increase in the amount of public information, by raising κ_y , will then raise the weight on the public signal even further, which is why more public information can reduce welfare.

With the equilibrium investment strategy in hand, I return to the firm's problem in Eq. (2).

Corollary 1. *The firm's optimal outside equity stake is $\lambda = \alpha$.*

Proof. See Appendix A.3. □

The founder trades off a larger share of the firm against the smaller investment that a larger share induces. Because equilibrium effective capital rises with the stake as $K_j \propto \lambda^{1/(1-\alpha)}$, the founder's payoff is proportional to $(1 - \lambda)\lambda^{\alpha/(1-\alpha)}$, which is maximized at $\lambda = \alpha$. The founder optimally cedes to investors a share equal to the output elasticity of the capital they provide.⁷

3.2. The Constrained Social Planner

To assess efficiency, I compare the equilibrium to the allocation chosen by a planner who faces the same informational friction as investors. The planner assigns each investor's contribution to each firm directly, but it cannot pool the information dispersed across investors: the contribution of investor i to firm j may depend only on that investor's information set $\omega_{i,j}$, that is, on the public signal y_j and the private signal $x_{i,j}$. Any difference between the two allocations therefore reflects how the available information is used, not how much information is available.

The planner chooses consumption and contributions to solve

$$\max_{c_1, c_2, \{k_{i,j}\}_{i \in I, j \in J}} \mathbb{E}[c_1 + \beta c_2] \tag{12}$$

⁷Equivalently, capital providers as a group receive a fraction α of output, the output elasticity of capital and hence the share capital would earn if it were paid its marginal product. It is somewhat reassuring that the equity contract reproduces this benchmark division of output between capital and the founder.

subject to

$$\begin{aligned} c_1 + \int_I \int_J k_{i,j} dj di &= e, \\ c_2 &= \int_J z_j K_j^\alpha dj + (1 - \delta) \int_I \int_J k_{i,j} dj di \end{aligned}$$

and to the constraint that $k_{i,j}$ be measurable with respect to $\omega_{i,j}$.

The first-order condition of Eq. (12) gives, for every $\omega_{i,j} \in \Omega$,

$$k^S(\omega_{i,j})^{1-\rho} = \frac{\alpha\beta}{1-\beta(1-\delta)} \mathbb{E}[z_j K_j^{\alpha-\rho} \mid \omega_{i,j}] \quad (13)$$

where $K_j = \left[\int_I k^S(\omega_{\ell,j})^\rho d\ell \right]^{1/\rho}$.

Applying to Eq. (13) the same scaling argument used for the decentralized equilibrium, a proportional increase in all other contributions by a factor $t > 0$ raises the planner's assignment by t^{θ^*} , where

$$\theta^* = \frac{\alpha - \rho}{1 - \rho} \quad (14)$$

I will refer to θ^* as the socially optimal degree of coordination.

Comparing Eq. (14) with Eq. (9), the two degrees of strategic complementarity are the same function of a single parameter, evaluated at ρ for the planner and at γ in the decentralized equilibrium. What differs is the force that offsets complementarity. For the planner it is the diminishing marginal product of an individual investor's capital in the CES aggregator, governed by ρ . In the decentralized equilibrium it is the dilution of an investor's share of the payout, governed by γ . Because $(\alpha - s)/(1 - s)$ is decreasing in s , we have $\theta > \theta^*$ precisely when $\gamma < \rho$. When shares respond less to own contributions than the technology warrants, investors coordinate more than is efficient.

The following proposition characterizes the planner's allocation.

Proposition 2. *The constrained efficient allocation is log-linear in the private and public signals:*

$$\ln k^S(\omega_{i,j}) = \phi_0^S + \phi_x^S x_{i,j} + \phi_y^S y_j$$

with loadings

$$\phi_x^S = \frac{1}{1 - \alpha} \cdot \frac{(1 - \theta^*)\kappa_x}{\kappa_z + \kappa_y + (1 - \theta^*)\kappa_x}, \quad \phi_y^S = \frac{1}{1 - \alpha} \cdot \frac{\kappa_y}{\kappa_z + \kappa_y + (1 - \theta^*)\kappa_x} \quad (15)$$

where θ^* is given by Eq. (14).

Proof. See Appendix A.4. □

The planner's allocation takes exactly the form of the equilibrium strategy in Eq. (10), with θ^* in place of θ . The two allocations therefore differ only through the coordination motive, which governs how investors weight their signals, and through the intercept, which governs the level of investment. This gives two conditions for efficiency.

Proposition 3. *The decentralized equilibrium attains the constrained efficient allocation if and only if*

(i) $\gamma = \rho$, and

(ii) $\tau = 1 - \gamma$.

Proof. See Appendix A.5. □

The two conditions correct two distinct distortions. Condition (i) aligns the coordination motives, setting $\theta = \theta^*$ so that investors weight their private and public signals exactly as the planner would. Appendix B.2 shows that $\gamma = \rho$ arises naturally in an environment where investors supply differentiated capital varieties and compete in quantities. Condition (ii) offsets investors' market power.

4 Welfare and the Social Value of Information

I now turn to the main question of the paper: when does improving investors' information about firm productivity raise welfare? Substituting the equilibrium allocation into the household's objective, welfare can be written as

$$\mathcal{W} = e - [1 - \beta(1 - \delta)]\bar{K} + \beta \mathcal{Y}, \quad \mathcal{Y} = \int_J z_j K_j^\alpha dj \quad (16)$$

where $\mathcal{Y}$ is aggregate output and $\bar{K}$ aggregate capital. Information affects welfare through these two aggregates. The following lemma describes the relationship between aggregate output and aggregate capital in this economy, and underlies the results that follow.

Lemma 2. *In equilibrium, aggregate capital is proportional to aggregate output, $\bar{K} = \chi \cdot \mathcal{Y}$, where $\chi = \gamma\alpha/[(1 - \tau)R]$ is the capital-output ratio, and welfare is*

$$\mathcal{W} = e + m\bar{K}, \quad m \equiv (1 - \beta(1 - \delta))\left(\frac{1 - \tau}{\gamma\alpha} - 1\right) \quad (17)$$

Proof. See Appendix A.6. □

Because aggregate output is proportional to aggregate capital, it is sufficient to focus on the effect of information on aggregate capital. Whether more information raises welfare then turns on how it moves aggregate capital and on the sign of m .

Information moves welfare in the same direction as capital when $m > 0$, and against it when $m < 0$. The sign of m is the sign of $1 - \tau - \gamma\alpha$: welfare rises with capital as long as the subsidy is not so generous that capital is over-supplied.⁸

The next subsection analyzes the constrained efficient case, in which more information of either kind is always beneficial. The two that follow move away from that benchmark, characterizing how welfare responds to the precision of the public and the private signal separately and showing that more of either can reduce welfare. The last subsection fixes the total amount of information available to investors and asks how it should be divided between public and private sources. Appendix E then extends the analysis to endogenous disclosure, letting firms choose how much public information to reveal.

4.1. Efficient Use of Information

When the conditions of Proposition 3 hold, the equilibrium coincides with the constrained efficient allocation for every information structure. More information is then unambiguously beneficial.

Proposition 4. *Suppose $\gamma = \rho$ and $\tau = 1 - \gamma$, so that the conditions of Proposition 3 hold. Then welfare is strictly increasing in the precision of both signals,*

$$\frac{\partial \mathcal{W}}{\partial \kappa_x} > 0 \quad \text{and} \quad \frac{\partial \mathcal{W}}{\partial \kappa_y} > 0$$

Proof. See Appendix A.7. □

Note first that under the efficient conditions $m = (1 - \beta(1 - \delta))(1 - \alpha)/\alpha > 0$, so by Lemma 2 welfare rises with aggregate capital. Therefore, the result follows from the fact that more information of either kind raises aggregate capital.

Proposition 4 is the counterpart, in this environment, of the efficient-use-of-information principle of Angeletos and Pavan (2007). They study a class of economies with dispersed information in which agents have quadratic payoffs over their own action, the aggregate action, and an unknown Gaussian fundamental. Within this class, more information of either kind raises welfare if two conditions hold. First, the equilibrium degree of coordination,

⁸When $1 - \tau < \gamma\alpha$ the subsidy induces over-investment: aggregate capital exceeds the welfare-maximizing level, so a further increase lowers welfare, and every sign in the results below reverses. Indeed, since $\mathcal{W} = e + m\bar{K}$ with $m < 0$, welfare falls short of e , the level the household would attain by consuming its entire endowment in period 1.

the elasticity of an agent's best response to the aggregate action, coincides with the degree a planner constrained to the same information would choose, so that information is used efficiently. Second, the decentralized equilibrium is efficient when the fundamental is common knowledge, so that no distortion operates independently of the informational friction. When either fails, more information of one kind or the other can instead reduce welfare.

The model presented in this paper can be read as a microfoundation of an economy of the kind they study. Its primitives are a CES technology, a profit-sharing rule, and log-normal productivity rather than an abstract quadratic payoff. The degree of strategic complementarity θ in Eq. (9) corresponds to their equilibrium degree of coordination and θ^* in Eq. (14) to its efficient counterpart, the objects they denote α and α^* .⁹ The two conditions above become the requirements of Proposition 3. The condition $\gamma = \rho$ delivers $\theta = \theta^*$ and equates the two degrees of coordination. As shown in Appendix C, the condition $\tau = 1 - \gamma$ is the one under which the equilibrium is efficient when productivities are common knowledge, for any ρ .¹⁰ In this sense Proposition 4 specializes their social value of information result to equity financing. Once information is used efficiently and market power is offset, more information of either kind raises welfare, and the result arises here from an explicit economic environment rather than a reduced-form payoff.

These conditions are restrictive. The parameter γ describes how a firm divides its payout among the investors that fund it, while ρ describes how their capital combines in production. There is no reason for the two to coincide. When they do not, $\theta \neq \theta^*$, information is used inefficiently, and the welfare effects of public and private information become ambiguous. The following two sections characterize those effects.

4.2. The Social Value of Public Information

Away from the benchmark of Proposition 4, more information need not raise welfare. I characterize the effect of public information here, and turn to the private signal next.

By Lemma 2, the welfare effect of the public signal is governed by its effect on aggregate

⁹Their α is the degree of coordination and is unrelated to the returns to capital α used in this paper.

¹⁰In fact, ρ plays no role when investors can observe the fundamental directly. As Appendix C shows, when all investors observe the fundamental their capital contributions are identical, and effective capital then equals that common contribution for any ρ . The substitutability of investors' capital matters only when their contributions differ, which requires that they observe different information. The same is true when private information is shut down ($\kappa_x \rightarrow 0$): with only the common prior and the public signal, investors again share the same information and contribute identically.

capital. As shown in Appendix A.7, aggregate capital can be written in closed form as

$$\bar{K} = \underbrace{\chi^{\frac{1}{1-\alpha}} \exp\left(\frac{\mu_z}{1-\alpha} + \frac{1}{2(1-\alpha)^2 \kappa_z}\right)}_{\bar{K}^{\text{FI}}} \cdot \underbrace{\exp\left(-\alpha \frac{(1-\alpha)(1-\rho)\kappa_x + (1-\gamma)^2(\kappa_y + \kappa_z)}{2(1-\alpha)^2 D^2}\right)}_{\text{information wedge}} \quad (18)$$

where $\chi = \gamma\alpha/[(1-\tau)R]$ and $D = (1-\gamma)(\kappa_y + \kappa_z) + (1-\alpha)\kappa_x$. The first factor is aggregate capital under full information, and the second is a wedge that captures the information friction.

Equation (18) has four implications. First, the wedge is always below one, so aggregate capital is always lower than under full information: dispersed information depresses investment. Second, the precisions κ_x and κ_y enter only through the wedge, and fiscal policy only through χ in the full-information factor, so the effect of information on aggregate capital is independent of fiscal policy. Third, the parameter ρ enters only through the private-signal term $(1-\rho)\kappa_x$. With no private information, $\kappa_x = 0$, and ρ drops out, since all investors then share the same information and contribute identically, and the CES aggregator returns that common contribution for any ρ . Fourth, as either signal becomes perfectly precise, $\kappa_x \rightarrow \infty$ or $\kappa_y \rightarrow \infty$, the wedge tends to one and aggregate capital approaches its full-information level, as expected.

The following proposition characterizes the effect of the precision of the public signal on aggregate capital.

Proposition 5. *Define the threshold*

$$\Delta_{\kappa_y} \equiv \frac{(1-\alpha)(2\rho - \gamma - 1)}{(1-\gamma)^2} \quad (19)$$

Then

$$\text{sgn}\left(\frac{\partial \bar{K}}{\partial \kappa_y}\right) = \text{sgn}\left(\frac{\kappa_y + \kappa_z}{\kappa_x} - \Delta_{\kappa_y}\right)$$

and the sign is the same for aggregate output.

Proof. See Appendix A.8. □

Combining Lemma 2 and Proposition 5 gives the welfare effect of the precision of the public signal.

Corollary 2. *The welfare effect of the precision of the public signal satisfies*

$$\text{sgn}\left(\frac{\partial \mathcal{W}}{\partial \kappa_y}\right) = \text{sgn}(m) \cdot \text{sgn}\left(\frac{\kappa_y + \kappa_z}{\kappa_x} - \Delta_{\kappa_y}\right)$$

In particular, when $m > 0$, welfare is increasing in the precision of the public signal if and only if $\frac{\kappa_y + \kappa_z}{\kappa_x} > \Delta_{\kappa_y}$.

Proof. It follows immediately from $\partial \mathcal{W} / \partial \kappa_y = m \partial \bar{K} / \partial \kappa_y$. $\square$

Take the region where $m > 0$. Two conditions must hold together for more public information to reduce welfare. First, the threshold must be positive, which by Eq. (19) requires

$$\rho > \frac{1 + \gamma}{2}, \quad \text{equivalently} \quad \theta > \frac{1 + \theta^*}{2}$$

The degree of strategic complementarity in the decentralized economy must exceed the midpoint between the socially optimal degree θ^* and one, so investors coordinate substantially more than is efficient. If instead $\rho \leq \frac{1 + \gamma}{2}$, then $\Delta_{\kappa_y} \leq 0$ and more public information always raises welfare. Second, public information must be imprecise relative to private information, $\frac{\kappa_y + \kappa_z}{\kappa_x} < \Delta_{\kappa_y}$.

When both conditions hold, a more precise public signal lowers welfare. The mechanism is the one identified by [Morris and Shin \(2002\)](#). Because the public signal is common knowledge, it is the natural instrument through which investors coordinate. When coordination is already excessive, a more precise public signal leads investors to rely on it even more heavily, as the loadings in Eq. (11) make explicit, amplifying the inefficient comovement of their investments and, through it, lowering aggregate capital and welfare.

The parallel with [Morris and Shin \(2002\)](#) is in fact exact. Written in terms of the equilibrium and the socially optimal degrees of coordination, θ and θ^* , the threshold of Proposition 5 becomes

$$\Delta_{\kappa_y} = \frac{(1 - \theta)(2\theta - 1 - \theta^*)}{1 - \theta^*}$$

so more public information raises welfare if and only if the ratio of public to private precision exceeds a threshold determined entirely by how much investors coordinate in equilibrium and how much the planner would have them coordinate.¹¹ This is the form of their central result. In their beauty-contest economy the coordination motive is a pure externality that cancels out on aggregation, so the socially optimal degree of coordination is zero. Setting $\theta^* = 0$, the threshold reduces to $(1 - \theta)(2\theta - 1)$, which is exactly their threshold, and the condition for public information to be potentially harmful becomes $\theta > \frac{1}{2}$, which is exactly their condition that the coordination weight exceed one half.

The comparison also clarifies what is new here. In their setting the payoff is quadratic and the socially optimal degree of coordination is zero by construction. Here both degrees come from primitives, θ from the returns to capital and the sharing rule and θ^* from the

¹¹The prior precision κ_z enters the public side of the ratio because the prior, like the public signal, is common to all investors. Morris and Shin assume an uninformative prior, so their ratio is κ_y / κ_x .

technology alone, and $\theta^* = 0$ corresponds to the knife-edge case $\rho = \alpha$. Their welfare result thus emerges as the special case of Corollary 2 in which the planner attaches no value to coordination, recovered from a microfounded equity-financing economy rather than from a reduced-form payoff.

Figure 1 partitions the (ρ, γ) plane into two regions according to the sign of the threshold: a blue region where $\Delta_{\kappa_y} \leq 0$ and more public information always raises welfare, and a red region where $\Delta_{\kappa_y} > 0$ and public information can be harmful. The locus of constrained efficiency $\gamma = \rho$ lies entirely in the blue region. Within the red region the threshold itself rises with ρ , shown by the darker shading: as investors' capital becomes more substitutable, the range of relative precisions over which public information is harmful widens.

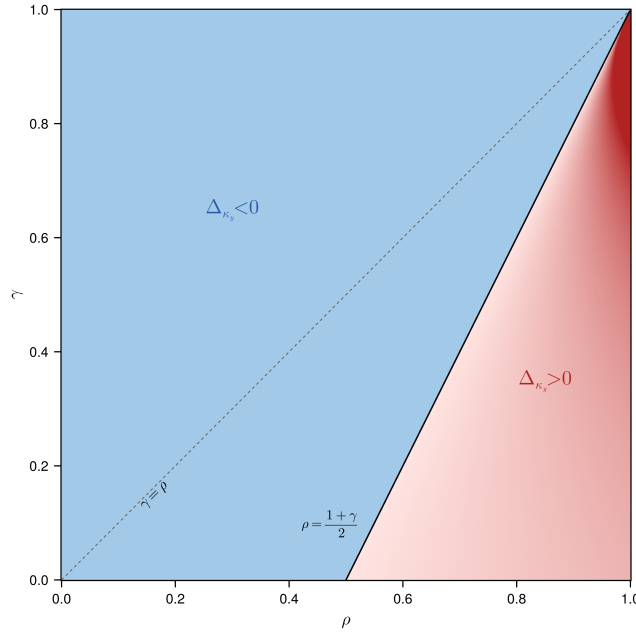


Figure 1: Sign and magnitude of the welfare threshold Δ_{κ_y}

Note: The figure partitions the (ρ, γ) plane by the sign of the threshold $\Delta_{\kappa_y} = \frac{(1-\alpha)(2\rho-\gamma-1)}{(1-\gamma)^2}$. In the blue region ($\rho < \frac{1+\gamma}{2}$) the threshold is negative and more precise public information always raises welfare. In the red region ($\rho > \frac{1+\gamma}{2}$) it is positive, and more precise public information reduces welfare when public information is sufficiently imprecise relative to private information, $\frac{\kappa_y + \kappa_z}{\kappa_x} < \Delta_{\kappa_y}$. Darker shading indicates a larger threshold. The dashed line $\gamma = \rho$ is the locus of constrained efficiency of Proposition 3.

4.3. The Social Value of Private Information

I now turn to the effect of more private information on welfare. As before, by Lemma 2, the welfare effect of the precision of the private signal is governed by its effect on aggregate capital, $\partial \bar{K} / \partial \kappa_x$, and its sign is independent of fiscal policy. Differentiating Eq. (18) with respect to κ_x

delivers a threshold characterization that mirrors Proposition 5.

Proposition 6. *Let $\rho < 1$ and define the threshold*

$$\Delta_{\kappa_x} \equiv \frac{(1 - \gamma)(2\gamma - \rho - 1)}{(1 - \alpha)(1 - \rho)} \quad (20)$$

Then

$$\text{sgn}\left(\frac{\partial \bar{K}}{\partial \kappa_x}\right) = \text{sgn}\left(\frac{\kappa_x}{\kappa_y + \kappa_z} - \Delta_{\kappa_x}\right)$$

and the sign is the same for aggregate output and for total factor productivity.

Proof. See Appendix A.9. □

Combining Lemma 2 and Proposition 6 gives the welfare effect of the precision of the private signal.

Corollary 3. *The welfare effect of the precision of the private signal satisfies*

$$\text{sgn}\left(\frac{\partial \mathcal{W}}{\partial \kappa_x}\right) = \text{sgn}(m) \text{sgn}\left(\frac{\kappa_x}{\kappa_y + \kappa_z} - \Delta_{\kappa_x}\right)$$

In particular, when $m > 0$, welfare is increasing in the precision of the private signal if and only if $\frac{\kappa_x}{\kappa_y + \kappa_z} > \Delta_{\kappa_x}$.

Proof. Immediate from $\partial \mathcal{W} / \partial \kappa_x = m \partial \bar{K} / \partial \kappa_x$. □

The structure mirrors Corollary 2, with the roles of γ and ρ interchanged. Taking again the region where $m > 0$, more private information reduces welfare if and only if two conditions hold together. First, the threshold must be positive, which by Eq. (20) requires

$$\gamma > \frac{1 + \rho}{2}, \quad \text{equivalently} \quad \theta^* > \frac{1 + \theta}{2}$$

Now it is the socially optimal degree of coordination that exceeds the midpoint between the degree of strategic complementarity θ and one, so investors coordinate substantially *less* than is efficient. Second, private information must be imprecise relative to public information,

$$\frac{\kappa_x}{\kappa_y + \kappa_z} < \Delta_{\kappa_x}.$$

The mechanism is the opposite of the one in Section 4.2. The private signal is the source of disagreement among investors: it is the only reason their information sets differ. When investors already coordinate too little, a more precise private signal disperses their contributions further, moving the allocation away from the one the planner would choose.

This result can again be compared with [Morris and Shin \(2002\)](#). Written in terms of the equilibrium and the socially optimal degrees of coordination, θ and θ^* , the threshold of Proposition 6 becomes

$$\Delta_{\kappa_x} = \frac{2\theta^* - \theta - 1}{(1 - \theta)^2}$$

In their economy more private information is always beneficial, and that result is nested here. Setting $\theta^* = 0$, the threshold becomes $-(1 + \theta)/(1 - \theta)^2$, which is negative, so by Corollary 3 welfare always rises with the precision of the private signal. Away from their benchmark the conclusion can reverse. Private information can be harmful only when $\theta^* > \frac{1+\theta}{2}$, that is, when the planner wants substantially more coordination than investors choose, something impossible in a beauty-contest economy where the socially optimal degree of coordination is zero. The possibility that private information reduces welfare is therefore an implication of the microfounded environment with no counterpart in their setting.

Figure 2 shows the two regions, and is the mirror image of Fig. 1. Private information can reduce welfare only in the red region, and the locus of constrained efficiency $\gamma = \rho$ again lies entirely in the blue region.

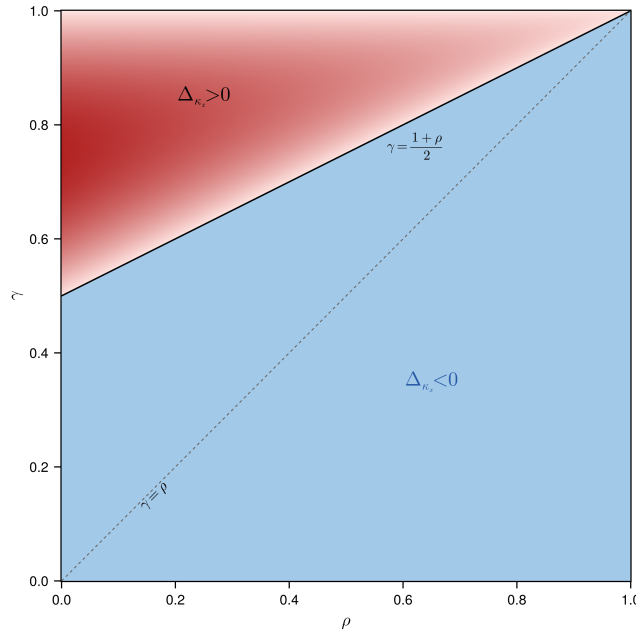


Figure 2: Sign and magnitude of the welfare threshold Δ_{κ_x}

Note: The figure partitions the (ρ, γ) plane by the sign of the threshold $\Delta_{\kappa_x} = \frac{(1-\gamma)(2\gamma-\rho-1)}{(1-\alpha)(1-\rho)}$. In the blue region ($\gamma < \frac{1+\rho}{2}$) the threshold is negative and more precise private information always raises welfare. In the red region ($\gamma > \frac{1+\rho}{2}$) it is positive, and more precise private information reduces welfare when private information is sufficiently imprecise relative to public information, $\frac{\kappa_x}{\kappa_y + \kappa_z} < \Delta_{\kappa_x}$. Darker shading indicates a larger threshold. The dashed line $\gamma = \rho$ is the locus of constrained efficiency of Proposition 3.

At $\gamma = \rho$ both thresholds are negative, $\Delta_{\kappa_y} = -\frac{1-\alpha}{1-\rho}$ and $\Delta_{\kappa_x} = -\frac{1-\rho}{1-\alpha}$, so more information of either kind raises welfare, as Proposition 4 requires. Moreover, the two harmful configurations are mutually exclusive: $\rho > \frac{1+\gamma}{2}$ and $\gamma > \frac{1+\rho}{2}$ cannot hold at once. In any given economy at most one of the two types of information can be harmful, and which one depends on whether the investors coordinate too much or too little.

4.4. The Optimal Composition of Information

The previous two sections took the precision of each signal as given. Following the approach of Angeletos, Iovino and La'O (2016), I now hold fixed the total amount of information available to investors and ask how it is best divided between the two sources. Put simply, given the total amount of information, should it be public or private?

Fix the total precision at κ and solve

$$\max_{\kappa_x, \kappa_y \geq 0} \mathcal{W} \quad \text{subject to} \quad \kappa_x + \kappa_y = \kappa$$

where $\mathcal{W}$ is welfare in the decentralized equilibrium. I focus here on the interesting case in which production is desirable in this economy, $m > 0$.

Proposition 7. *Let $m > 0$ and $\kappa > 0$. If $\gamma = \rho = \alpha$, welfare does not depend on the composition of information and any division of κ is optimal. Otherwise, the welfare-maximizing information structure is always a corner: either fully public, $\kappa_y = \kappa$, or fully private, $\kappa_x = \kappa$. It is fully public if*

$$\rho < \alpha + \frac{\kappa_z(\gamma - \alpha)^2}{(1 - \alpha)(\kappa + \kappa_z)} \quad (21)$$

and fully private if the reverse strict inequality holds. Either corner is optimal when Eq. (21) holds with equality.

Proof. See Appendix A.10. □

The first part of the result states that, outside the knife-edge case $\gamma = \rho = \alpha$, dividing the available information between the two sources is never optimal. Along the constraint $\kappa_x + \kappa_y = \kappa$, welfare has at most one interior critical point, and that point is a minimum rather than a maximum. In the knife-edge case investors have no incentive to coordinate their actions, $\theta = 0$, and this coincides with what the planner would want, $\theta^* = 0$. The composition of information is then simply irrelevant.

The second part states which corner is optimal outside the knife-edge case. The right-hand side of Eq. (21) is at least α , so information should be fully public whenever $\rho \leq \alpha$, that is, whenever $\theta^* \geq 0$. When the planner would want investors to coordinate, information should

be public, regardless of the amount of information available or the precision of the prior. When instead $\rho > \alpha$, so that $\theta^* < 0$ and the planner would want investors not to coordinate, information should be private once ρ is far enough above α . The gap between the two cutoffs is the correction term in Eq. (21), which vanishes when $\gamma = \alpha$, so that $\theta = 0$ and investors have no strategic motive, and which shrinks as κ grows. In the limit of abundant information the rule is simply that information should be public if and only if $\theta^* > 0$.

Figure 3 plots the two regions.

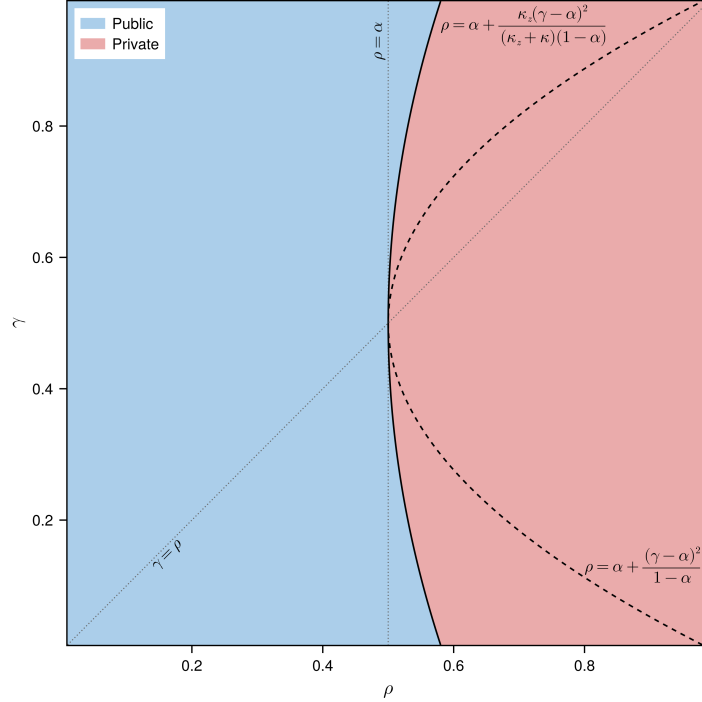


Figure 3: Optimal composition of information

Note: The figure partitions the (ρ, γ) plane according to whether welfare is higher when all information is public or when all of it is private. The solid curve is the boundary Eq. (21): information should be fully public in the blue region and fully private in the red region. The dotted vertical line is $\rho = \alpha$, the boundary in the limit of abundant information $\kappa \rightarrow \infty$ or when the prior is uninformative $\kappa_z = 0$. The dashed curve is $\rho = \alpha + (\gamma - \alpha)^2 / (1 - \alpha)$, the boundary in the opposite limit $\kappa \rightarrow 0$ or when $\kappa_z \rightarrow \infty$. The solid curve lies between the two. The dotted diagonal is $\gamma = \rho$.

5 Conclusion

This paper asks when more information about firms raises welfare when firms are financed with equity. I study an economy in which many investors fund each firm without observing its productivity, relying instead on a private and a public signal. How much investors coordinate their contributions depends on how substitutable their capital is in production and on how strongly an investor's share of the payout rises with its own contribution. When the share

responds to a contribution exactly as it would if investors were paid their marginal product, and a subsidy offsets investors' market power, the equilibrium is constrained efficient and more information of either kind raises welfare. Otherwise, investors coordinate too much when an investor's share responds weakly to its own contribution relative to how substitutable their capital is, and too little in the opposite case. More public information can then lower welfare when they coordinate too much, and more private information can lower welfare when they coordinate too little. Finally, a policymaker who can decide only whether information is public or private, not how much there is, should make it either fully public or fully private. Information should be fully public whenever coordination among investors is socially desirable.

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A Omitted Proofs

A.1. Proof of Lemma 1

The argument follows [Skaperdas \(1996\)](#), adapted to a continuum of investors. Throughout I drop the firm subscript j and write k_i for the contribution of investor i .

Proof. By separability, part (iii) of Assumption 1, there is a continuous and increasing function f with

$$s_i = \frac{f(k_i)}{\int_I f(k_\ell) d\ell} \quad (22)$$

For the shares to be well defined and nonnegative for every profile, as feasibility requires, f must be strictly positive. Feasibility is then automatic: the shares in Eq. (22) are positive and integrate to one by construction. It remains to use scale invariance to pin down f .

Fix a profile and a scalar $t > 0$, and scale every contribution by t . By part (ii) of Assumption 1, each investor’s share is unchanged, so for any two investors i and ℓ ,

$$\frac{f(tk_i)}{\int_I f(tk_m) dm} = \frac{f(k_i)}{\int_I f(k_m) dm}, \quad \frac{f(tk_\ell)}{\int_I f(tk_m) dm} = \frac{f(k_\ell)}{\int_I f(k_m) dm}$$

The common denominators cancel when the two are divided, leaving

$$\frac{f(tk_i)}{f(k_i)} = \frac{f(tk_\ell)}{f(k_\ell)}$$

Because the profile was arbitrary, this holds for every pair of contributions. Hence the ratio $f(tx)/f(x)$ does not depend on x . Evaluating it at $x = 1$ and writing x for a generic contribution,

$$\frac{f(tx)}{f(x)} = \frac{f(t)}{f(1)} \iff f(tx)f(1) = f(t)f(x) \quad (23)$$

Define $g(x) \equiv f(x)/f(1)$, which is continuous and strictly positive. Dividing Eq. (23) by

$f(1)^2$ gives the multiplicative Cauchy equation

$$g(tx) = g(t)g(x) \quad \text{for all } t, x > 0$$

Substituting $x = e^u$ and $t = e^v$ and setting $G(u) \equiv \ln g(e^u)$, which is well defined and continuous because g is continuous and positive, this becomes the additive Cauchy equation $G(u + v) = G(u) + G(v)$ for all $u, v \in \mathbb{R}$. Its only continuous solutions are linear, $G(u) = \gamma u$ for some $\gamma \in \mathbb{R}$, so that $g(x) = x^\gamma$ and

$$f(x) = f(1)x^\gamma$$

Substituting into Eq. (22), the constant $f(1)$ cancels between numerator and denominator and

$$s_i = \frac{k_i^\gamma}{\int_I k_\ell^\gamma d\ell}$$

which is Eq. (3). Finally, f is increasing, so $\gamma > 0$. □

A.2. Proof of Proposition 1

Proof. Throughout I drop the firm subscript j and write k_i , x_i , y and z . It is convenient to define

$$\chi \equiv \frac{\gamma \lambda}{(1 - \tau)R}, \quad S \equiv \kappa_x + \kappa_y + \kappa_z, \quad D \equiv (1 - \gamma)(\kappa_y + \kappa_z) + (1 - \alpha)\kappa_x$$

so that the fixed point in Eq. (6) becomes $k_i^{1-\gamma} = \chi \mathbb{E}\left[zK^\alpha / \int_I k_\ell^\gamma d\ell \mid \omega_{i,j}\right]$. At the firm's optimal stake $\lambda = \alpha$ of Corollary 1, χ is the capital-output ratio of Lemma 2. I guess that the equilibrium strategy is log-linear in the signals,

$$\ln k_i = \phi_0 + \phi_x x_i + \phi_y y$$

and verify that coefficients (ϕ_0, ϕ_x, ϕ_y) exist that make the guess consistent with the fixed point.

Raising the guess to the power ρ and integrating,

$$\begin{aligned} \int_I k_\ell^\rho d\ell &= \exp(\rho(\phi_0 + \phi_y y)) \int_I \exp(\rho\phi_x x_\ell) d\ell \\ &= \exp(\rho(\phi_0 + \phi_y y + \phi_x \ln z)) \int_I \exp(\rho\phi_x v_\ell) d\ell \\ &= \exp\left(\rho(\phi_0 + \phi_y y + \phi_x \ln z) + \frac{(\rho\phi_x)^2}{2\kappa_x}\right) \end{aligned}$$

where the second line substitutes $x_\ell = \ln z + v_\ell$ and the third applies a law of large numbers

across the continuum of investors: the noise $v_\ell \sim \mathcal{N}(0, 1/\kappa_x)$ is independent across ℓ , so $\int_I \exp(\rho \phi_x v_\ell) d\ell = \mathbb{E}[\exp(\rho \phi_x v)] = \exp((\rho \phi_x)^2 / 2\kappa_x)$. Raising to the power α/ρ ,

$$K^\alpha = \left[\int_I k_\ell^\rho d\ell \right]^{\alpha/\rho} = \exp \left(\alpha \left(\phi_0 + \phi_y y + \phi_x \ln z \right) + \frac{\alpha \rho \phi_x^2}{2\kappa_x} \right)$$

The same computation with γ in place of ρ gives the denominator of the sharing rule,

$$\int_I k_\ell^\gamma d\ell = \exp \left(\gamma \left(\phi_0 + \phi_y y + \phi_x \ln z \right) + \frac{(\gamma \phi_x)^2}{2\kappa_x} \right)$$

Given $\omega_{i,j} = (x_i, y)$, Gaussian updating gives

$$\mathbb{E}[\ln z \mid \omega_{i,j}] = \frac{\kappa_z \mu_z + \kappa_x x_i + \kappa_y y}{S}, \quad \text{Var}(\ln z \mid \omega_{i,j}) = \frac{1}{S}$$

Both $\ln K^\alpha$ and $\ln \int_I k_\ell^\gamma d\ell$ are linear functions of $\ln z$, so conditional on $\omega_{i,j}$ each is normally distributed. Their means are

$$\begin{aligned} \mathbb{E}[\ln K^\alpha \mid \omega_{i,j}] &= \alpha \phi_0 + \alpha \phi_y y + \frac{\alpha \rho \phi_x^2}{2\kappa_x} + \alpha \phi_x \mathbb{E}[\ln z \mid \omega_{i,j}], \\ \mathbb{E} \left[\ln \int_I k_\ell^\gamma d\ell \mid \omega_{i,j} \right] &= \gamma \phi_0 + \gamma \phi_y y + \frac{(\gamma \phi_x)^2}{2\kappa_x} + \gamma \phi_x \mathbb{E}[\ln z \mid \omega_{i,j}] \end{aligned}$$

and their variances and covariances are

$$\begin{aligned} \text{Var}(\ln K^\alpha \mid \omega_{i,j}) &= \frac{(\alpha \phi_x)^2}{S}, & \text{Cov}(\ln z, \ln K^\alpha \mid \omega_{i,j}) &= \frac{\alpha \phi_x}{S}, \\ \text{Var} \left(\ln \int_I k_\ell^\gamma d\ell \mid \omega_{i,j} \right) &= \frac{(\gamma \phi_x)^2}{S}, & \text{Cov} \left(\ln z, \ln \int_I k_\ell^\gamma d\ell \mid \omega_{i,j} \right) &= \frac{\gamma \phi_x}{S} \end{aligned}$$

together with

$$\text{Cov} \left(\ln K^\alpha, \ln \int_I k_\ell^\gamma d\ell \mid \omega_{i,j} \right) = \alpha \gamma \phi_x^2 \text{Var}(\ln z \mid \omega_{i,j}) = \frac{\alpha \gamma \phi_x^2}{S}$$

Define:

$$W \equiv \ln z + \ln K^\alpha - \ln \int_I k_\ell^\gamma d\ell$$

so that the expectation in the fixed point is $\mathbb{E}[\exp W \mid \omega_{i,j}]$. Since W is conditionally normal, $\mathbb{E}[\exp W \mid \omega_{i,j}] = \exp(\mathbb{E}[W \mid \omega_{i,j}] + \frac{1}{2} \text{Var}(W \mid \omega_{i,j}))$. Collecting the means above,

$$\mathbb{E}[W \mid \omega_{i,j}] = [1 + (\alpha - \gamma) \phi_x] \mathbb{E}[\ln z \mid \omega_{i,j}] + (\alpha - \gamma) \phi_0 + (\alpha - \gamma) \phi_y y + \frac{(\alpha \rho - \gamma^2) \phi_x^2}{2\kappa_x}$$

For the variance, $\mathbb{V}\text{ar}(W \mid \omega_{i,j})$ equals the sum of the three variances plus $2\mathbb{C}\text{ov}(\ln z, \ln K^\alpha) - 2\mathbb{C}\text{ov}(\ln z, \ln \int_I k_\ell^\gamma) - 2\mathbb{C}\text{ov}(\ln K^\alpha, \ln \int_I k_\ell^\gamma)$, all conditional on $\omega_{i,j}$, which gives

$$\mathbb{V}\text{ar}(W \mid \omega_{i,j}) = \frac{1 + (\alpha\phi_x)^2 + (\gamma\phi_x)^2 + 2\alpha\phi_x - 2\gamma\phi_x - 2\alpha\gamma\phi_x^2}{S} = \frac{[1 + (\alpha - \gamma)\phi_x]^2}{S}$$

Taking logs of the fixed point and substituting the guess,

$$(1 - \gamma)(\phi_0 + \phi_x x_i + \phi_y y) = \ln \chi + \mathbb{E}[W \mid \omega_{i,j}] + \frac{1}{2} \mathbb{V}\text{ar}(W \mid \omega_{i,j})$$

Matching coefficients on x_i gives

$$(1 - \gamma)\phi_x = [1 + (\alpha - \gamma)\phi_x] \frac{\kappa_x}{S} \implies \phi_x [(1 - \gamma)S - (\alpha - \gamma)\kappa_x] = \kappa_x$$

and since $(1 - \gamma)S - (\alpha - \gamma)\kappa_x = D$,

$$\phi_x = \frac{\kappa_x}{D} \quad (24)$$

It follows that

$$1 + (\alpha - \gamma)\phi_x = \frac{D + (\alpha - \gamma)\kappa_x}{D} = \frac{(1 - \gamma)S}{D} \quad (25)$$

Matching coefficients on y gives $(1 - \gamma)\phi_y = [1 + (\alpha - \gamma)\phi_x]\kappa_y/S + (\alpha - \gamma)\phi_y$, so that $(1 - \alpha)\phi_y = [1 + (\alpha - \gamma)\phi_x]\kappa_y/S$, and using Eq. (25),

$$\phi_y = \frac{(1 - \gamma)\kappa_y}{(1 - \alpha)D} \quad (26)$$

Finally, matching the constant terms,

$$(1 - \gamma)\phi_0 = \ln \chi + [1 + (\alpha - \gamma)\phi_x] \frac{\kappa_z \mu_z}{S} + (\alpha - \gamma)\phi_0 + \frac{(\alpha\rho - \gamma^2)\phi_x^2}{2\kappa_x} + \frac{[1 + (\alpha - \gamma)\phi_x]^2}{2S}$$

Collecting the terms in ϕ_0 on the left and substituting Eq. (24) and Eq. (25),

$$(1 - \alpha)\phi_0 = \ln \chi + \frac{(1 - \gamma)\kappa_z \mu_z}{D} + \frac{\kappa_x(\alpha\rho - \gamma^2)}{2D^2} + \frac{(1 - \gamma)^2 S}{2D^2}$$

so that

$$\phi_0 = \frac{1}{1 - \alpha} \left[\ln \chi + \frac{(1 - \gamma)\kappa_z \mu_z}{D} + \frac{\kappa_x(\alpha\rho - \gamma^2) + (1 - \gamma)^2 S}{2D^2} \right] \quad (27)$$

Since $1 - \theta = (1 - \alpha)/(1 - \gamma)$ by Eq. (9), we have $D = (1 - \gamma)[\kappa_z + \kappa_y + (1 - \theta)\kappa_x]$, and therefore

$$\phi_x = \frac{1}{1 - \alpha} \cdot \frac{(1 - \theta)\kappa_x}{\kappa_z + \kappa_y + (1 - \theta)\kappa_x}, \quad \phi_y = \frac{1}{1 - \alpha} \cdot \frac{\kappa_y}{\kappa_z + \kappa_y + (1 - \theta)\kappa_x}$$

which are the loadings stated in Eq. (11). The coefficients (ϕ_0, ϕ_x, ϕ_y) are uniquely determined, so the initial guess was correct. $\square$

A.3. Proof of Corollary 1

Proof. Consider a firm choosing its stake λ , anticipating that the investors who fund it play the equilibrium strategy of Proposition 1. The stake enters that strategy only through the constant $\chi = \gamma\lambda/[(1 - \tau)R]$, and hence only through the intercept. From Eq. (27),

$$\phi_0 = \frac{\ln \lambda}{1 - \alpha} + \text{terms independent of } \lambda$$

while the loadings Eq. (24) and Eq. (26) do not involve λ at all. Every contribution to the firm is therefore proportional to $\lambda^{1/(1-\alpha)}$, and so is effective capital,

$$K_j(\lambda) = \lambda^{\frac{1}{1-\alpha}} \tilde{K}_j$$

where $\tilde{K}_j$ does not depend on λ .

Substituting into Eq. (2), and noting that $\lambda^{\alpha/(1-\alpha)}$ is a constant that factors out of the expectation, the founder solves

$$\max_{\lambda \in [0,1]} (1 - \lambda) \lambda^{\frac{\alpha}{1-\alpha}} \mathbb{E}[z_j \tilde{K}_j^\alpha]$$

which is equivalent to maximizing $(1 - \lambda)\lambda^{\alpha/(1-\alpha)}$. It is easy to see that $\lambda = \alpha$ maximizes this expression. $\square$

A.4. Proof of Proposition 2

Proof. Throughout, $\rho < 1$. Substituting the two constraints of Eq. (12) directly into the objective, the planner solves

$$\max \mathbb{E} \left[e - \int_I \int_J k_{i,j} dj di + \beta \left(\int_J z_j \left(\int_I k_{i,j}^\rho di \right)^{\alpha/\rho} dj + (1 - \delta) \int_I \int_J k_{i,j} dj di \right) \right]$$

where the choice of $k_{i,j}$ must be a function of $x_{i,j}$ and y_j only. The first-order condition with respect to $k_{i,j}$ is

$$\mathbb{E}\left[-1 + \beta \left(z_j \frac{\alpha}{\rho} \left(\int_I k_{i,j}^\rho di \right)^{\frac{\alpha}{\rho}-1} \rho k_{i,j}^{\rho-1} + (1-\delta) \right)\right] = 0$$

Since $k_{i,j}^{\rho-1}$ is a function of $x_{i,j}$ and y_j , it can be taken outside the expectation, which leaves

$$\beta \alpha \mathbb{E}\left[z_j \left(\int_I k_{i,j}^\rho di \right)^{\frac{\alpha}{\rho}-1} \middle| \omega_{i,j} \right] k_{i,j}^{\rho-1} = 1 - \beta(1-\delta)$$

that is,

$$k_{i,j}^{1-\rho} = \frac{\beta \alpha}{1 - \beta(1-\delta)} \mathbb{E}\left[z_j \left(\int_I k_{i,j}^\rho di \right)^{\frac{\alpha}{\rho}-1} \middle| \omega_{i,j} \right]$$

Writing $K_j^\rho = \int_I k_{i,j}^\rho di$, so that $\left(\int_I k_{i,j}^\rho di \right)^{\alpha/\rho-1} = K_j^{\alpha-\rho}$,

$$k^S(\omega_{i,j})^{1-\rho} = \frac{\beta \alpha}{1 - \beta(1-\delta)} \mathbb{E}\left[z_j K_j^{\alpha-\rho} \middle| \omega_{i,j} \right]$$

which is Eq. (13). The condition depends on i only through $\omega_{i,j}$, so the planner's allocation is symmetric, as in the decentralized equilibrium.

By the definition of the CES aggregator, $\int_I k^S(\omega_{\ell,j})^\rho d\ell = K_j^\rho$, so that

$$z_j K_j^{\alpha-\rho} = \frac{z_j K_j^\alpha}{\int_I k^S(\omega_{\ell,j})^\rho d\ell}$$

Using $\beta \alpha / [1 - \beta(1-\delta)] = \alpha/R$, the planner's condition can be written

$$k^S(\omega_{i,j})^{1-\rho} = \frac{\alpha}{R} \mathbb{E}\left[\frac{z_j K_j^\alpha}{\int_I k^S(\omega_{\ell,j})^\rho d\ell} \middle| \omega_{i,j} \right]$$

This is exactly the equilibrium fixed point Eq. (6), with ρ in place of γ and with the constant $\chi = \gamma \lambda / [(1-\tau)R]$ replaced by α/R . The derivation in Appendix A.2 nowhere assumes $\gamma \neq \rho$, so it applies also to the fixed point above.

Writing $S = \kappa_x + \kappa_y + \kappa_z$ and

$$D^S \equiv (1-\rho)(\kappa_y + \kappa_z) + (1-\alpha)\kappa_x$$

Eq. (24), Eq. (26) and Eq. (27) evaluated at $\gamma = \rho$ and $\chi = \alpha/R$ give $\ln k^S(\omega_{i,j}) = \phi_0^S + \phi_x^S x_{i,j} + \phi_y^S y_j$

with

$$\phi_x^S = \frac{\kappa_x}{D^S}, \quad \phi_y^S = \frac{(1-\rho)\kappa_y}{(1-\alpha)D^S}$$

and

$$\phi_0^S = \frac{1}{1-\alpha} \left[\ln \frac{\alpha}{R} + \frac{(1-\rho)\kappa_z \mu_z}{D^S} + \frac{\kappa_x \rho (\alpha - \rho) + (1-\rho)^2 S}{2(D^S)^2} \right]$$

where the term $\alpha\rho - \gamma^2$ in Eq. (27) becomes $\alpha\rho - \rho^2 = \rho(\alpha - \rho)$.

Since $1 - \theta^* = (1 - \alpha)/(1 - \rho)$ by Eq. (14), we have $D^S = (1 - \rho)[\kappa_z + \kappa_y + (1 - \theta^*)\kappa_x]$, and therefore

$$\phi_x^S = \frac{1}{1-\alpha} \cdot \frac{(1-\theta^*)\kappa_x}{\kappa_z + \kappa_y + (1-\theta^*)\kappa_x}, \quad \phi_y^S = \frac{1}{1-\alpha} \cdot \frac{\kappa_y}{\kappa_z + \kappa_y + (1-\theta^*)\kappa_x}$$

which are the loadings in Eq. (15). □

A.5. Proof of Proposition 3

Proof. Both the equilibrium strategy and the planner's allocation are log-linear in the two signals with uniquely determined coefficients, so the two allocations coincide if and only if all three coefficients coincide. Write $D = (1-\gamma)(\kappa_y + \kappa_z) + (1-\alpha)\kappa_x$ and $D^S = (1-\rho)(\kappa_y + \kappa_z) + (1-\alpha)\kappa_x$ as before.

From Eq. (24) and Appendix A.4, $\phi_x = \kappa_x/D$ and $\phi_x^S = \kappa_x/D^S$, so $\phi_x = \phi_x^S$ if and only if $D = D^S$. Since

$$D - D^S = (\rho - \gamma)(\kappa_y + \kappa_z)$$

and $\kappa_y + \kappa_z > 0$, this holds if and only if $\gamma = \rho$. Equivalently, $\theta = \theta^*$. When $\gamma = \rho$ the two denominators coincide and, comparing Eq. (26) with $\phi_y^S = (1-\rho)\kappa_y/[(1-\alpha)D^S]$, the loadings on the public signal coincide as well.

Suppose $\gamma = \rho$. Comparing Eq. (27) with the intercept in Appendix A.4, what remains is to show that $\phi_0 = \phi_0^S$. That is equivalent to $\chi = \alpha/R$,

$$\frac{\gamma\lambda}{(1-\tau)R} = \frac{\alpha}{R} \iff \frac{\gamma\lambda}{1-\tau} = \alpha$$

Using $\lambda = \alpha$ from Corollary 1, this reduces to $1 - \tau = \gamma$.

The decentralized equilibrium therefore attains the constrained efficient allocation if and only if $\gamma = \rho$ and $\tau = 1 - \gamma$. □

A.6. Proof of Lemma 2

Proof. Return to the fixed point Eq. (6) and write $P_j \equiv \int_I k_{\ell,j}^\gamma d\ell$ for the denominator of the sharing rule. Multiplying both sides by $k_{i,j}^\gamma$, which is measurable with respect to $\omega_{i,j}$ and so can be taken inside the expectation,

$$k_{i,j} = \chi \mathbb{E} \left[\frac{k_{i,j}^\gamma z_j K_j^\alpha}{P_j} \middle| \omega_{i,j} \right]$$

where $\chi = \frac{\gamma\lambda}{(1-\tau)R}$.

Taking unconditional expectations on both sides and applying the law of iterated expectations,

$$\mathbb{E}[k_{i,j}] = \chi \mathbb{E} \left[\frac{k_{i,j}^\gamma z_j K_j^\alpha}{P_j} \right]$$

Now condition on (z_j, y_j) . Both K_j and P_j are functions of (z_j, y_j) alone, while a law of large numbers across the continuum of investors gives $\mathbb{E}[k_{i,j}^\gamma | z_j, y_j] = P_j$. The denominator therefore cancels,

$$\mathbb{E} \left[\frac{k_{i,j}^\gamma z_j K_j^\alpha}{P_j} \right] = \mathbb{E} \left[\frac{z_j K_j^\alpha}{P_j} \mathbb{E}[k_{i,j}^\gamma | z_j, y_j] \right] = \mathbb{E}[z_j K_j^\alpha] = \mathcal{Y}$$

where the last equality is a law of large numbers across firms. Since the same argument applied to the left-hand side gives $\mathbb{E}[k_{i,j}] = \bar{K}$, aggregate capital is proportional to aggregate output,

$$\bar{K} = \chi \mathcal{Y} \tag{28}$$

Substituting Eq. (28) into Eq. (16) and using $\beta R = 1 - \beta(1 - \delta)$,

$$\begin{aligned} \mathcal{W} &= e + \beta \mathcal{Y} - \bar{K} [1 - \beta(1 - \delta)] \\ &= e + \bar{K} (1 - \beta(1 - \delta)) \left(\frac{1 - \tau}{\gamma\lambda} - 1 \right) \end{aligned}$$

Substituting $\lambda = \alpha$ from Corollary 1 gives Eq. (17). □

A.7. Proof of Proposition 4

Proof. By Lemma 2, welfare is a linear function of aggregate capital, so it suffices to derive $\bar{K}$ in closed form and sign its derivatives. I compute $\bar{K}$ first, then impose the conditions of Proposition 3 and sign the effect of each precision.

As in Appendix A.2, write

$$\chi \equiv \frac{\gamma\lambda}{(1-\tau)R}, \quad S \equiv \kappa_x + \kappa_y + \kappa_z, \quad D \equiv (1-\gamma)(\kappa_y + \kappa_z) + (1-\alpha)\kappa_x$$

and recall from Eq. (24), Eq. (26) and Eq. (27) that $\phi_x = \kappa_x/D$, $\phi_y = (1-\gamma)\kappa_y/[(1-\alpha)D]$ and

$$\phi_0 = \frac{1}{1-\alpha} \left[\ln \chi + \frac{(1-\gamma)\kappa_z\mu_z}{D} + \frac{\kappa_x(\alpha\rho - \gamma^2) + (1-\gamma)^2 S}{2D^2} \right]$$

Integrating the equilibrium strategy $k_{i,j} = \exp(\phi_0 + \phi_x x_{i,j} + \phi_y y_j)$ across investors and firms,

$$\begin{aligned} \bar{K} &= \int_I \int_I \exp(\phi_0 + \phi_x x_{i,j} + \phi_y y_j) di dj \\ &= \exp\left(\phi_0 + \frac{\phi_x^2}{2\kappa_x}\right) \int_I \exp(\phi_y y_j + \phi_x \ln z_j) dj \\ &= \exp\left(\phi_0 + \frac{\phi_x^2}{2\kappa_x}\right) \mathbb{E}[\exp((\phi_x + \phi_y) \ln z_j)] \mathbb{E}[\exp(\phi_y \varepsilon_j)] \end{aligned}$$

The second line integrates over investors, using $x_{i,j} = \ln z_j + v_{i,j}$ and a law of large numbers across the continuum: $\int_I \exp(\phi_x v_{i,j}) di = \exp(\phi_x^2/2\kappa_x)$. The third line substitutes $y_j = \ln z_j + \varepsilon_j$, applies a law of large numbers across firms, and uses the independence of $\ln z_j$ and ε_j to factor the expectation. Evaluating the two log-normal moments,

$$\bar{K} = \exp \left(\underbrace{\phi_0 + \frac{\phi_x^2}{2\kappa_x} + \frac{\phi_y^2}{2\kappa_y} + (\phi_x + \phi_y)\mu_z + \frac{(\phi_x + \phi_y)^2}{2\kappa_z}}_{\equiv E} \right) \quad (29)$$

It remains to simplify E . Since $D - (1-\gamma)\kappa_z = (1-\gamma)\kappa_y + (1-\alpha)\kappa_x$, the loadings satisfy

$$\phi_x + \phi_y = \frac{(1-\alpha)\kappa_x + (1-\gamma)\kappa_y}{(1-\alpha)D} = \frac{1}{1-\alpha} \left[1 - \frac{(1-\gamma)\kappa_z}{D} \right] \quad (30)$$

Substituting ϕ_0 and Eq. (30) into E , the two terms in $\kappa_z\mu_z$ cancel, leaving $\mu_z/(1-\alpha)$. Using $\phi_x^2/(2\kappa_x) = \kappa_x/(2D^2)$ and $\phi_y^2/(2\kappa_y) = (1-\gamma)^2\kappa_y/[2(1-\alpha)^2D^2]$, and expanding the square in Eq. (30),

$$\frac{(\phi_x + \phi_y)^2}{2\kappa_z} = \frac{1}{2(1-\alpha)^2\kappa_z} - \frac{1-\gamma}{(1-\alpha)^2D} + \frac{(1-\gamma)^2\kappa_z}{2(1-\alpha)^2D^2}$$

so that

$$E = \frac{\ln \chi}{1-\alpha} + \frac{\mu_z}{1-\alpha} + \frac{1}{2(1-\alpha)^2 \kappa_z} + \frac{N}{2(1-\alpha)^2 D^2}$$

where, after placing the remaining terms over the common denominator $2(1-\alpha)^2 D^2$,

$$N = (1-\alpha) \left[\kappa_x (\alpha \rho - \gamma^2) + (1-\gamma)^2 S \right] + (1-\alpha)^2 \kappa_x + (1-\gamma)^2 \kappa_y + (1-\gamma)^2 \kappa_z - 2(1-\gamma)D$$

Grouping the terms in κ_x and in $\kappa_y + \kappa_z$ separately, and using $S = \kappa_x + \kappa_y + \kappa_z$ and $D = (1-\gamma)(\kappa_y + \kappa_z) + (1-\alpha)\kappa_x$,

$$\begin{aligned} N &= (1-\alpha)\kappa_x \left[(\alpha \rho - \gamma^2) + (1-\gamma)^2 + (1-\alpha) - 2(1-\gamma) \right] + (1-\gamma)^2 (\kappa_y + \kappa_z) [(1-\alpha) + 1 - 2] \\ &= -\alpha \left[(1-\alpha)(1-\rho)\kappa_x + (1-\gamma)^2 (\kappa_y + \kappa_z) \right] \end{aligned}$$

where the bracket multiplying $(1-\alpha)\kappa_x$ collapses because $(\alpha \rho - \gamma^2) + (1-2\gamma + \gamma^2) + (1-\alpha) - 2 + 2\gamma = \alpha \rho - \alpha$. Substituting E into Eq. (29) gives the closed form for aggregate capital stated in Eq. (18).

By Lemma 2, welfare is $\mathcal{W} = e + m\bar{K}$ with $m = (1 - \beta(1 - \delta)) \left(\frac{1-\tau}{\gamma\alpha} - 1 \right)$. Impose the conditions of Proposition 3, that is, $\rho = \gamma$ and $1 - \tau = \gamma$. The policy condition gives $(1 - \tau)/(\gamma\alpha) = 1/\alpha$, so that

$$\mathcal{W} = e + \frac{(1 - \beta(1 - \delta))(1 - \alpha)}{\alpha} \bar{K} \quad (31)$$

and welfare is strictly increasing in aggregate capital. It therefore suffices to sign the effect of each precision on $\bar{K}$.

The first condition, $\rho = \gamma$, simplifies the exponent of Eq. (18) to:

$$(1-\alpha)(1-\rho)\kappa_x + (1-\gamma)^2 (\kappa_y + \kappa_z) = (1-\gamma) \left[(1-\alpha)\kappa_x + (1-\gamma)(\kappa_y + \kappa_z) \right] = (1-\gamma)D$$

so that Eq. (18) collapses to

$$\bar{K} = \chi^{\frac{1}{1-\alpha}} \exp \left(\frac{\mu_z}{1-\alpha} + \frac{1}{2(1-\alpha)^2 \kappa_z} \right) \exp \left(-\frac{\alpha(1-\gamma)}{2(1-\alpha)^2 D} \right) \quad (32)$$

Only the last factor depends on κ_x and κ_y , and it is strictly increasing in D . Since $\partial D / \partial \kappa_x = 1 - \alpha > 0$ and $\partial D / \partial \kappa_y = 1 - \gamma > 0$, aggregate capital is strictly increasing in the precision of both signals. Combining this with Eq. (31) gives $\partial \mathcal{W} / \partial \kappa_x > 0$ and $\partial \mathcal{W} / \partial \kappa_y > 0$. $\square$

A.8. Proof of Proposition 5

Proof. From Eq. (18), and since the constant χ does not depend on the precisions, $\ln \bar{K}$ depends on the precisions only through the exponent of the last factor,

$$\ln \bar{K} = \text{const} + E, \quad E = -\frac{\alpha}{2(1-\alpha)^2} \cdot \frac{(1-\alpha)(1-\rho)\kappa_x + (1-\gamma)^2(\kappa_y + \kappa_z)}{D^2}$$

where $D = (1-\gamma)(\kappa_y + \kappa_z) + (1-\alpha)\kappa_x$. Since the omitted terms do not depend on κ_y , $\text{sgn}(\partial \bar{K} / \partial \kappa_y) = \text{sgn}(\partial E / \partial \kappa_y)$.

Differentiate E . Write $H \equiv (1-\alpha)(1-\rho)\kappa_x + (1-\gamma)^2(\kappa_y + \kappa_z)$ for the numerator, so that $\partial H / \partial \kappa_y = (1-\gamma)^2$ and $\partial D / \partial \kappa_y = 1-\gamma$. By the quotient rule,

$$\frac{\partial}{\partial \kappa_y} \left(\frac{H}{D^2} \right) = \frac{(1-\gamma)^2 D^2 - H \cdot 2D(1-\gamma)}{D^4} = \frac{1-\gamma}{D^3} [(1-\gamma)D - 2H]$$

We can simplify the bracket:

$$\begin{aligned} (1-\gamma)D - 2H &= (1-\gamma) \left[(1-\gamma)(\kappa_y + \kappa_z) + (1-\alpha)\kappa_x \right] - 2 \left[(1-\alpha)(1-\rho)\kappa_x + (1-\gamma)^2(\kappa_y + \kappa_z) \right] \\ &= (1-\alpha)(2\rho - \gamma - 1)\kappa_x - (1-\gamma)^2(\kappa_y + \kappa_z) \end{aligned}$$

Therefore

$$\frac{\partial E}{\partial \kappa_y} = -\frac{\alpha(1-\gamma)}{2(1-\alpha)^2 D^3} \left[(1-\alpha)(2\rho - \gamma - 1)\kappa_x - (1-\gamma)^2(\kappa_y + \kappa_z) \right]$$

The leading fraction is negative, so $\partial \bar{K} / \partial \kappa_y > 0$ if and only if the bracket is negative, that is,

$$(1-\gamma)^2(\kappa_y + \kappa_z) > (1-\alpha)(2\rho - \gamma - 1)\kappa_x \iff \frac{\kappa_y + \kappa_z}{\kappa_x} > \Delta_{\kappa_y}$$

where $\Delta_{\kappa_y} \equiv (1-\alpha)(2\rho - \gamma - 1)/(1-\gamma)^2$ is the threshold in Proposition 5. Finally, Lemma 2 gives $\mathcal{Y} = \bar{K}/\chi$, and total factor productivity is $\mathcal{Y}/\bar{K}^\alpha = \bar{K}^{1-\alpha}/\chi$. Both are strictly increasing transformations of $\bar{K}$, so their derivatives with respect to κ_y share its sign. $\square$

A.9. Proof of Proposition 6

Proof. The argument follows Appendix A.8, differentiating with respect to κ_x rather than κ_y . With E , H and D as defined there, $\partial H / \partial \kappa_x = (1-\alpha)(1-\rho)$ and $\partial D / \partial \kappa_x = 1-\alpha$, so the quotient rule gives

$$\frac{\partial}{\partial \kappa_x} \left(\frac{H}{D^2} \right) = \frac{(1-\alpha)(1-\rho)D^2 - H \cdot 2D(1-\alpha)}{D^4} = \frac{1-\alpha}{D^3} [(1-\rho)D - 2H]$$

Expanding the bracket and collecting the terms in $\kappa_y + \kappa_z$ and in κ_x ,

$$\begin{aligned}(1 - \rho)D - 2H &= (1 - \rho) \left[(1 - \gamma)(\kappa_y + \kappa_z) + (1 - \alpha)\kappa_x \right] - 2 \left[(1 - \alpha)(1 - \rho)\kappa_x + (1 - \gamma)^2(\kappa_y + \kappa_z) \right] \\ &= (1 - \gamma)(2\gamma - \rho - 1)(\kappa_y + \kappa_z) - (1 - \alpha)(1 - \rho)\kappa_x\end{aligned}$$

Therefore,

$$\frac{\partial E}{\partial \kappa_x} = -\frac{\alpha}{2(1 - \alpha)D^3} \left[(1 - \gamma)(2\gamma - \rho - 1)(\kappa_y + \kappa_z) - (1 - \alpha)(1 - \rho)\kappa_x \right]$$

The leading fraction is negative, so $\partial \bar{K} / \partial \kappa_x > 0$ if and only if the bracket is negative, that is,

$$(1 - \alpha)(1 - \rho)\kappa_x > (1 - \gamma)(2\gamma - \rho - 1)(\kappa_y + \kappa_z) \iff \frac{\kappa_x}{\kappa_y + \kappa_z} > \Delta_{\kappa_x}$$

where $\Delta_{\kappa_x} \equiv (1 - \gamma)(2\gamma - \rho - 1) / [(1 - \alpha)(1 - \rho)]$ is the threshold in Proposition 6. The extension to aggregate output and total factor productivity is as in Appendix A.8. $\square$

A.10. Proof of Proposition 7

Proof. By Lemma 2 with $m > 0$, maximizing welfare is equivalent to maximizing $\bar{K}$, hence to maximizing the exponent in Eq. (18),

$$E = -\frac{\alpha}{2(1 - \alpha)^2} \cdot \frac{N}{D^2}, \quad \begin{aligned} N &= (1 - \alpha)(1 - \rho)\kappa_x + (1 - \gamma)^2(\kappa_y + \kappa_z), \\ D &= (1 - \gamma)(\kappa_y + \kappa_z) + (1 - \alpha)\kappa_x \end{aligned}$$

Since the leading constant is negative, maximizing E is equivalent to minimizing N/D^2 .

Substituting $\kappa_x = \kappa - \kappa_y$, so that $d\kappa_x/d\kappa_y = -1$,

$$\frac{d}{d\kappa_y} \left(\frac{N}{D^2} \right) = \frac{D(N_y - N_x) - 2N(D_y - D_x)}{D^3}$$

where subscripts denote partial derivatives, $N_y = (1 - \gamma)^2$, $N_x = (1 - \alpha)(1 - \rho)$, $D_y = 1 - \gamma$ and $D_x = 1 - \alpha$, so that $D_y - D_x = \alpha - \gamma$. Since $D^3 > 0$, the sign is that of

$$D \left[(1 - \gamma)^2 - (1 - \alpha)(1 - \rho) \right] - 2(\alpha - \gamma)N$$

Collect the terms in $\kappa_y + \kappa_z$. Factoring out $1 - \gamma$,

$$\begin{aligned}
& (\kappa_y + \kappa_z)(1 - \gamma) \left[(1 - \gamma)^2 - (1 - \alpha)(1 - \rho) - 2(\alpha - \gamma)(1 - \gamma) \right] \\
&= (\kappa_y + \kappa_z)(1 - \gamma) \left[(1 - \gamma)(1 + \gamma - 2\alpha) - (1 - \alpha)(1 - \rho) \right] \\
&= (\kappa_y + \kappa_z)(1 - \gamma) \left[(1 - \alpha)^2 - (\alpha - \gamma)^2 - (1 - \alpha)(1 - \rho) \right] \\
&= -(\kappa_y + \kappa_z)(1 - \gamma) B_1
\end{aligned}$$

where the third line uses $(1 - \gamma)(1 + \gamma - 2\alpha) = (1 - \alpha)^2 - (\alpha - \gamma)^2$ and

$$B_1 \equiv (\alpha - \gamma)^2 + (1 - \alpha)(\alpha - \rho)$$

Collect next the terms in κ_x . Factoring out $1 - \alpha$,

$$\begin{aligned}
& (1 - \alpha)\kappa_x \left[(1 - \gamma)^2 - (1 - \alpha)(1 - \rho) - 2(\alpha - \gamma)(1 - \rho) \right] \\
&= (1 - \alpha)\kappa_x \left[1 - 2\gamma\rho + \gamma^2 - (1 + \alpha)(1 - \rho) \right] \\
&= (1 - \alpha)\kappa_x \left[(\gamma - \rho)^2 + (1 - \rho)(1 + \rho) - (1 + \alpha)(1 - \rho) \right] \\
&= (1 - \alpha)\kappa_x B_2
\end{aligned}$$

where the third line uses $1 - 2\gamma\rho + \gamma^2 = (\gamma - \rho)^2 + (1 - \rho)(1 + \rho)$ and

$$B_2 \equiv (\gamma - \rho)^2 - (1 - \rho)(\alpha - \rho)$$

Since welfare moves opposite to N/D^2 , there is a positive constant such that

$$\left. \frac{dW}{d\kappa_y} \right|_{\kappa} = \text{constant} \times \left[(\kappa_y + \kappa_z)(1 - \gamma)B_1 - \kappa_x(1 - \alpha)B_2 \right] \quad (33)$$

First, B_1 and B_2 cannot both be nonpositive unless both are zero: $B_1 \leq 0$ requires $(1 - \alpha)(\alpha - \rho) \leq -(\alpha - \gamma)^2 \leq 0$ and hence $\alpha \leq \rho$, while $B_2 \leq 0$ requires $(1 - \rho)(\alpha - \rho) \geq (\gamma - \rho)^2 \geq 0$ and hence $\alpha \geq \rho$. Both nonpositive therefore forces $\alpha = \rho$, and then $B_1 \leq 0$ reads $(\alpha - \gamma)^2 \leq 0$, so $\gamma = \alpha$ and $B_1 = B_2 = 0$. Three cases remain: both zero, exactly one strictly positive, or both strictly positive.

Suppose first $B_1 = B_2 = 0$, which by the argument above happens exactly when $\gamma = \rho = \alpha$. The bracket in Eq. (33) then vanishes identically, so welfare does not depend on the split.

Indeed, substituting $\gamma = \rho = \alpha$ and $\kappa_x = \kappa - \kappa_y$ into the exponent,

$$\begin{aligned} E &= -\alpha \frac{(1-\rho)^2 \left[(\kappa - \kappa_y) + (\kappa_y + \kappa_z) \right]}{2(1-\alpha)^2(1-\rho)^2 \left[(\kappa_y + \kappa_z) + (\kappa - \kappa_y) \right]^2} \\ &= -\frac{\alpha}{2(1-\alpha)^2(\kappa + \kappa_z)} \end{aligned}$$

for every $\kappa_y \in [0, \kappa]$. Any composition of information then maximizes welfare, and in particular so does either corner.

If $B_1 > 0$ and $B_2 \leq 0$, both terms in Eq. (33) are nonnegative and the first is strictly positive, so welfare is strictly increasing in κ_y and the optimum is $\kappa_y = \kappa$. If $B_1 \leq 0$ and $B_2 > 0$, both terms are nonpositive and the second is strictly negative, so welfare is strictly decreasing in κ_y and the optimum is $\kappa_y = 0$.

If $B_1 > 0$ and $B_2 > 0$, substitute $\kappa_x = \kappa - \kappa_y$ in Eq. (33) to obtain

$$\left. \frac{dW}{d\kappa_y} \right|_{\kappa} = \text{constant} \times \left[\kappa_y ((1-\gamma)B_1 + (1-\alpha)B_2) + \kappa_z(1-\gamma)B_1 - \kappa(1-\alpha)B_2 \right]$$

which is linear in κ_y with strictly positive slope. The derivative is therefore negative to the left of its unique zero and positive to the right, so that interior critical point is a minimum of welfare, not a maximum. Therefore, in every case a corner is optimal.

Evaluate E at each corner. With $\kappa_x = 0$ and $\kappa_y = \kappa$, $N = (1-\gamma)^2(\kappa + \kappa_z)$ and $D = (1-\gamma)(\kappa + \kappa_z)$, so

$$E_{\text{public}} = \frac{-\alpha}{2(1-\alpha)^2(\kappa + \kappa_z)}$$

while with $\kappa_x = \kappa$ and $\kappa_y = 0$,

$$E_{\text{private}} = -\alpha \frac{(1-\alpha)(1-\rho)\kappa + (1-\gamma)^2\kappa_z}{2(1-\alpha)^2[(1-\gamma)\kappa_z + (1-\alpha)\kappa]^2}$$

Dividing by the negative constant $-\alpha/[2(1-\alpha)^2]$ reverses the inequality, so $E_{\text{public}} > E_{\text{private}}$ if and only if

$$\left[(1-\alpha)(1-\rho)\kappa + (1-\gamma)^2\kappa_z \right] (\kappa + \kappa_z) > [(1-\gamma)\kappa_z + (1-\alpha)\kappa]^2$$

Expanding both sides,

$$(1-\alpha)(\alpha-\rho)\kappa + \kappa_z \left[(1-\gamma)^2 + (1-\alpha)(1-\rho) - 2(1-\gamma)(1-\alpha) \right] > 0$$

The bracket equals B_1 , since $(1-\gamma)^2 - 2(1-\gamma)(1-\alpha) = (\alpha-\gamma)^2 - (1-\alpha)^2$. The condition is

therefore

$$(1 - \alpha)(\alpha - \rho)(\kappa + \kappa_z) + \kappa_z(\alpha - \gamma)^2 > 0$$

which rearranges to Eq. (21), and the two corners attain the same welfare exactly when Eq. (21) holds with equality. $\square$

B Microfoundations and Alternative Environments

B.1. Taste-Shock Microfoundation

The sharing rule was derived axiomatically in Appendix A.1. This appendix gives a behavioral microfoundation: the same rule arises when the founder allocates equity through a discrete-choice problem with idiosyncratic taste shocks. Because a choice among a continuum of alternatives is not well defined, I state the mechanism for a firm funded by a finite number N of investors and recover the continuum as the limit $N \rightarrow \infty$. I drop the firm subscript j for clarity.

The firm's equity is a continuum of infinitesimal claims. When the founder allocates a given claim, it assigns to investor i a perceived value

$$V_i = \eta f(k_i) + \varepsilon_i, \quad \varepsilon_i \stackrel{iid}{\sim} \text{Gumbel}$$

where f is continuous and increasing and $\eta > 0$ weights the deterministic valuation against the taste shock. The claim is awarded to the investor with the highest perceived value, and the shocks are drawn independently across claims. The ε_i capture idiosyncratic frictions in how the founder matches with investors, such as personal ties or timing, that are unrelated to the capital provided. The share of equity that investor i receives is the probability that it attains the highest perceived value, which for Gumbel shocks is the familiar logit form

$$s_i = \frac{\exp(\eta f(k_i))}{\sum_{\ell=1}^N \exp(\eta f(k_\ell))} \quad (34)$$

Set $F(k) \equiv \exp(\eta f(k))$, which is continuous, strictly positive, and increasing because f is. In terms of F the logit share Eq. (34) is $s_i = F(k_i) / \sum_{\ell=1}^N F(k_\ell)$, which is exactly the separable form Eq. (22) with F in place of f . Feasibility and separability, parts (i) and (iii) of Assumption 1, therefore hold automatically. Imposing scale invariance, part (ii), leads to the same functional equation for F as the one solved in the proof of Lemma 1, namely Eq. (23) with F in place of f , whose continuous solutions are the power functions $F(k) = F(1)k^\gamma$ for some $\gamma > 0$. The constant $F(1)$ cancels in the share, leaving $s_i = k_i^\gamma / \sum_{\ell=1}^N k_\ell^\gamma$.

In the continuum each investor is one of infinitely many, so any individual's share is zero

and the relevant object is the density of equity across investors. Indexing the N investors by an evenly spaced grid on $I = [0, 1]$, the density assigned to investor i is N times its share,

$$Ns_i = \frac{k_i^\gamma}{\frac{1}{N} \sum_{\ell=1}^N k_\ell^\gamma}$$

As $N \rightarrow \infty$ the average in the denominator converges to $\int_I k_\ell^\gamma d\ell$. Restoring the firm subscript, the density converges to

$$s_{i,j} = \frac{k_{i,j}^\gamma}{\int_I k_{\ell,j}^\gamma d\ell}$$

which is the sharing rule Eq. (3) and integrates to one across investors.

Two remarks. First, the extreme-value assumption is not incidental: it is what makes the choice probabilities multiplicative in $\exp(\eta f)$, so that F is separable and the argument of Lemma 1 applies. A different distribution of taste shocks would deliver a different sharing rule. Second, $F(k) = F(1)k^\gamma$ means $f(k) = a \ln k + b$ with $\gamma = \eta a$, so γ can be interpreted as a signal-to-noise ratio. When the deterministic valuation dominates the taste shocks, η is large, γ is large, and shares respond sharply to contributions. When taste shocks dominate, η and hence γ are small, the allocation is close to random, and the rule approaches the equal division discussed in Section 2.3.

B.2. Isomorphism with Monopolistic Competition

The sharing rule of Lemma 1 was derived axiomatically. This appendix shows that the particular case $\gamma = \rho$, under which the equilibrium uses information efficiently by Proposition 3, arises on its own in a familiar environment: one in which investors supply differentiated varieties of capital and compete in quantities.

Environment. Investor i supplies a distinct variety of capital to firm j , which the firm rents at a price $P_{i,j}$ per unit and combines with the other varieties using the same CES technology as in Eq. (1). There is no equity contract and no sharing rule: investors are compensated through prices. The timing is as in the baseline. Investors choose quantities in period 1, before productivity is realized, and each bears a cost of funds $R(1 - \tau)$ per unit. In period 2 productivity is realized, the firm produces, and the firm pays each investor out of output. Prices are not agreed in advance. $P_{i,j}$ is the realized marginal product of variety i , so it varies with z_j and investors continue to bear the firm's productivity risk.

The firm's problem. Firm j chooses how much of each variety to rent,

$$\max_{\{k_{i,j}\}} z_j \left[\int_I k_{i,j}^\rho di \right]^{\alpha/\rho} - \int_I P_{i,j} k_{i,j} di$$

The first-order condition with respect to $k_{i,j}$ is $z_j \frac{\alpha}{\rho} \left[\int_I k_{\ell,j}^\rho d\ell \right]^{\frac{\alpha}{\rho}-1} \rho k_{i,j}^{\rho-1} = P_{i,j}$. Writing $K_j = \left[\int_I k_{\ell,j}^\rho d\ell \right]^{1/\rho}$ as before, the inverse demand curve for variety i is

$$P_{i,j} = \alpha z_j K_j^{\alpha-\rho} k_{i,j}^{\rho-1} \quad (35)$$

Two usual properties of Eq. (35) are worth noting. First, demand for each variety is downward sloping, with elasticity $1/(1-\rho)$, so an investor that supplies more depresses the price it receives. Second, integrating Eq. (35) against $k_{i,j}$ and using $\int_I k_{\ell,j}^\rho d\ell = K_j^\rho$,

$$\int_I P_{i,j} k_{i,j} di = \alpha z_j K_j^{\alpha-\rho} \int_I k_{i,j}^\rho di = \alpha z_j K_j^\alpha \quad (36)$$

The firm pays out exactly the fraction α of its output and keeps $(1-\alpha)z_j K_j^\alpha$.

The investor's problem. Investor i chooses a quantity, taking the inverse demand Eq. (35) into account and, having measure zero, taking K_j as given. Its problem is

$$\max_{k_{i,j}} \mathbb{E}[P_{i,j} k_{i,j} - R(1-\tau)k_{i,j} \mid \omega_{i,j}] \quad \text{subject to} \quad P_{i,j} = \alpha z_j K_j^{\alpha-\rho} k_{i,j}^{\rho-1}$$

Substituting the constraint and using $K_j^{\alpha-\rho} = K_j^\alpha / K_j^\rho$ together with $K_j^\rho = \int_I k_{\ell,j}^\rho d\ell$,

$$\begin{aligned} \mathbb{E}[\alpha z_j K_j^{\alpha-\rho} k_{i,j}^{\rho-1} k_{i,j} - R(1-\tau)k_{i,j} \mid \omega_{i,j}] &= \mathbb{E}[\alpha z_j K_j^{\alpha-\rho} k_{i,j}^\rho - R(1-\tau)k_{i,j} \mid \omega_{i,j}] \\ &= \mathbb{E} \left[\underbrace{\alpha}_{(1)} \underbrace{z_j K_j^\alpha}_{(2)} \underbrace{\frac{k_{i,j}^\rho}{\int_I k_{\ell,j}^\rho d\ell}}_{(3)} - R(1-\tau)k_{i,j} \mid \omega_{i,j} \right] \end{aligned}$$

The isomorphism. The last line is exactly the investor's problem of the baseline model, Eq. (4). Term by term: (1) is the fraction of output accruing to investors, which equals the outside equity stake λ , and by Eq. (36) it equals α , the value that solves the firm's problem in the baseline, (2) is the firm's output, and (3) is the sharing rule Eq. (3) evaluated at $\gamma = \rho$. The differentiated-varieties economy is therefore isomorphic to the baseline model with $\lambda = \alpha$ and

$\gamma = \rho$.

The economics behind $\gamma = \rho$ is that the same parameter now plays both roles. In the baseline, ρ governs how an investor's capital contributes to production while γ governs how it is rewarded, and the two are unrelated. Here the reward is a price, and the price is the marginal product, so the elasticity of an investor's compensation with respect to its own contribution must equal the elasticity of output with respect to that contribution.

C The Full-Information Benchmark

This appendix solves the model when firm productivity is common knowledge, so that every investor observes z_j . I first solve the decentralized equilibrium, and then turn to the planner's allocation.

C.1. Decentralized Equilibrium

Investor i solves

$$\max_{k_{i,j}} \frac{k_{i,j}^\gamma}{\int_I k_{\ell,j}^\gamma d\ell} \lambda z_j K_j^\alpha - R(1 - \tau) k_{i,j}, \quad K_j = \left[\int_I k_{\ell,j}^\rho d\ell \right]^{1/\rho}$$

The first-order condition for this problem is:

$$k_{i,j}^{1-\gamma} = \frac{\gamma \lambda}{R(1 - \tau)} \frac{z_j K_j^\alpha}{\int_I k_{\ell,j}^\gamma d\ell}$$

The right-hand side does not depend on i , so every investor funding firm j contributes the same amount and the subscript i can be dropped. Setting $k_{i,j} = k_j$ gives $K_j = k_j$ and $\int_I k_{\ell,j}^\gamma d\ell = k_j^\gamma$, so the condition collapses to $\gamma \lambda z_j / [R(1 - \tau)] = k_j^{1-\alpha}$, that is,

$$k_j = \left[\frac{\gamma \lambda z_j}{R(1 - \tau)} \right]^{\frac{1}{1-\alpha}} \quad (37)$$

Note that ρ drops out: with symmetric contributions the CES aggregator returns k_j for any ρ .

Aggregates. Productivity is independent across a continuum of firms, so cross-sectional averages equal expectations. With $\ln z_j \sim \mathcal{N}(\mu_z, 1/\kappa_z)$,

$$\int_J z_j^{\frac{1}{1-\alpha}} dj = \mathbb{E} \left[z_j^{\frac{1}{1-\alpha}} \right] = \exp \left(\frac{\mu_z}{1-\alpha} + \frac{1}{2(1-\alpha)^2 \kappa_z} \right)$$

Aggregating Eq. (37), and using $K_j = k_j$:

$$\bar{K} = \int_J \int_I k_{i,j} di dj = \left[\frac{\gamma \lambda}{R(1-\tau)} \right]^{\frac{1}{1-\alpha}} \exp\left(\frac{\mu_z}{1-\alpha} + \frac{1}{2(1-\alpha)^2 \kappa_z} \right), \quad (38)$$

$$\mathcal{Y} = \int_J z_j K_j^\alpha dj = \left[\frac{\gamma \lambda}{R(1-\tau)} \right]^{\frac{\alpha}{1-\alpha}} \exp\left(\frac{\mu_z}{1-\alpha} + \frac{1}{2(1-\alpha)^2 \kappa_z} \right) \quad (39)$$

Total factor productivity. Defining TFP = $\mathcal{Y}/\bar{K}^\alpha$, the terms in $\gamma \lambda/[R(1-\tau)]$ cancel and

$$\text{TFP} = \exp\left(\mu_z + \frac{1}{2(1-\alpha)\kappa_z} \right) \quad (40)$$

This is the highest level of total factor productivity attainable in this economy. It depends on neither the sharing rule nor policy: γ , λ and τ move the level of investment but not its allocation across firms, which is proportional to $z_j^{1/(1-\alpha)}$.

Welfare. Substituting Eq. (38) and Eq. (39) into Eq. (16), and using $1 - \beta(1 - \delta) = \beta R$,

$$\begin{aligned} \mathcal{W} &= e + \beta \exp\left(\frac{\mu_z}{1-\alpha} + \frac{1}{2(1-\alpha)^2 \kappa_z} \right) \left[\frac{\beta \gamma \lambda}{(1 - \beta(1 - \delta))(1 - \tau)} \right]^{\frac{\alpha}{1-\alpha}} \left(1 - \frac{\gamma \lambda}{1 - \tau} \right) \\ &= e + \beta \exp\left(\frac{\mu_z}{1-\alpha} + \frac{1}{2(1-\alpha)^2 \kappa_z} \right) \left[\frac{\beta \gamma \alpha}{(1 - \beta(1 - \delta))(1 - \tau)} \right]^{\frac{\alpha}{1-\alpha}} \left(1 - \frac{\gamma \alpha}{1 - \tau} \right) \end{aligned} \quad (41)$$

where in the last line I substitute $\lambda = \alpha$ from the firm's problem.

C.2. Social Planner

Under full information the planner chooses how much capital each investor $i \in I$ provides to each firm $j \in J$ so as to maximize welfare directly,

$$\max_{\{k_{i,j}\}_{i \in I, j \in J}} \mathcal{W} = e - (1 - \beta(1 - \delta)) \int_J \int_I k_{i,j} di dj + \beta \int_J z_j \left[\int_I k_{i,j}^\rho di \right]^{\alpha/\rho} dj$$

The first-order condition with respect to $k_{i,j}$ is

$$1 - \beta(1 - \delta) = \beta \alpha z_j \left[\int_I k_{\ell,j}^\rho d\ell \right]^{\frac{\alpha}{\rho}-1} k_{i,j}^{\rho-1}$$

As in the decentralized equilibrium, the right-hand side is the same for every investor funding firm j , so the subscript can be dropped and $k_{i,j} = k_j$. Substituting, $\left[k_j^\rho \right]^{\frac{\alpha}{\rho}-1} k_j^{\rho-1} = k_j^{\alpha-1}$, so the

condition becomes $1 - \beta(1 - \delta) = \beta\alpha z_j k_j^{\alpha-1}$, that is,

$$k_j^S = \left[\frac{\beta\alpha z_j}{1 - \beta(1 - \delta)} \right]^{\frac{1}{1-\alpha}} \quad (42)$$

Here too ρ drops out, for the same reason as in Eq. (37).

Aggregates. Aggregating Eq. (42) exactly as before,

$$\begin{aligned} \bar{K}^S &= \left[\frac{\beta\alpha}{1 - \beta(1 - \delta)} \right]^{\frac{1}{1-\alpha}} \exp\left(\frac{\mu_z}{1 - \alpha} + \frac{1}{2(1 - \alpha)^2 \kappa_z} \right), \\ \mathcal{Y}^S &= \left[\frac{\beta\alpha}{1 - \beta(1 - \delta)} \right]^{\frac{\alpha}{1-\alpha}} \exp\left(\frac{\mu_z}{1 - \alpha} + \frac{1}{2(1 - \alpha)^2 \kappa_z} \right) \end{aligned}$$

Total factor productivity. The terms in $\beta\alpha/[1 - \beta(1 - \delta)]$ again cancel in $\mathcal{Y}^S/(\bar{K}^S)^\alpha$, so

$$\text{TFP}^S = \exp\left(\mu_z + \frac{1}{2(1 - \alpha)\kappa_z} \right)$$

which is exactly Eq. (40).

Welfare. Substituting into Eq. (16) and simplifying as in Eq. (41),

$$\mathcal{W}^S = e + \beta(1 - \alpha) \exp\left(\frac{\mu_z}{1 - \alpha} + \frac{1}{2(1 - \alpha)^2 \kappa_z} \right) \left[\frac{\beta\alpha}{1 - \beta(1 - \delta)} \right]^{\frac{\alpha}{1-\alpha}} \quad (43)$$

Implementation. Comparing Eq. (37) with Eq. (42), the two allocations coincide for every z_j if and only if

$$\frac{\gamma\lambda}{1 - \tau} = \alpha$$

Using $\lambda = \alpha$ from Corollary 1, this reduces to $\tau = 1 - \gamma$. The same condition equates Eq. (41) and Eq. (43), since it makes the last factor of Eq. (41) equal to $1 - \alpha$.

Note that ρ plays no role: under full information the only distortion is the level of investment, which arises from investors' market power, and the subsidy $\tau = 1 - \gamma$ removes it. The condition $\gamma = \rho$ that Proposition 3 also requires is not needed here. It governs how investors weight their signals rather than how much they invest, and so it binds only when information is dispersed.

D Relaxing Risk Neutrality

This appendix shows that the welfare results of Section 4 do not depend on the household being risk neutral, provided that $\tau = 1 - \gamma$. Let the household have preferences $U(c_1, c_2)$, increasing and concave, and solve

$$\max_{c_1, c_2, \bar{K}} U(c_1, c_2)$$

subject to $c_1 + \bar{K} \leq e$ and $c_2 + T \leq (1 - \delta + R)\bar{K} + \Pi^F + \Pi^I$. Nothing else in the model changes: the investors' problem takes the cost of capital R as given, so Proposition 1, Corollary 1 and Lemma 2 continue to hold. Only the determination of R is affected.

Letting λ_1 and λ_2 be the multipliers on the two constraints, the first-order conditions for c_1 , c_2 and $\bar{K}$ are $U_1 = \lambda_1$, $U_2 = \lambda_2$ and $\lambda_2(1 - \delta + R) = \lambda_1$. Writing $R^* \equiv R + 1 - \delta$ for the gross return on savings,

$$U_1(c_1, c_2) = R^* U_2(c_1, c_2) \quad (44)$$

Under risk neutrality $U_1/U_2 = 1/\beta$ and Eq. (44) reduces to Eq. (8). Substituting the resource constraints, welfare is

$$\mathcal{W} = U(e - \bar{K}, \mathcal{Y} + (1 - \delta)\bar{K})$$

Differentiating with respect to κ_y and using Eq. (44),

$$\begin{aligned} \frac{\partial \mathcal{W}}{\partial \kappa_y} &= -U_1 \frac{\partial \bar{K}}{\partial \kappa_y} + U_2 \left[\frac{\partial \mathcal{Y}}{\partial \kappa_y} + (1 - \delta) \frac{\partial \bar{K}}{\partial \kappa_y} \right] \\ &= U_2 \left[-R^* \frac{\partial \bar{K}}{\partial \kappa_y} + (1 - \delta) \frac{\partial \bar{K}}{\partial \kappa_y} + \frac{\partial \mathcal{Y}}{\partial \kappa_y} \right] = U_2 \left[-R \frac{\partial \bar{K}}{\partial \kappa_y} + \frac{\partial \mathcal{Y}}{\partial \kappa_y} \right] \end{aligned}$$

since $-R^* + (1 - \delta) = -R$.

Write Γ for total factor productivity, so that $\mathcal{Y} = \Gamma \bar{K}^\alpha$ and

$$\frac{\partial \mathcal{Y}}{\partial \kappa_y} = \frac{\partial \Gamma}{\partial \kappa_y} \bar{K}^\alpha + \alpha \frac{\mathcal{Y}}{\bar{K}} \frac{\partial \bar{K}}{\partial \kappa_y} = \frac{\partial \Gamma}{\partial \kappa_y} \bar{K}^\alpha + \alpha \frac{R(1 - \tau)}{\gamma \lambda} \frac{\partial \bar{K}}{\partial \kappa_y}$$

where the second equality uses $\mathcal{Y}/\bar{K} = 1/\chi$ from Lemma 2. Substituting and imposing $\lambda = \alpha$ from Corollary 1,

$$\frac{\partial \mathcal{W}}{\partial \kappa_y} = U_2 \left[R \left(\frac{1 - \tau}{\gamma} - 1 \right) \frac{\partial \bar{K}}{\partial \kappa_y} + \frac{\partial \Gamma}{\partial \kappa_y} \bar{K}^\alpha \right] \quad (45)$$

The first term is the investment channel and the second the productivity channel.

Setting $1 - \tau = \gamma$ makes the first term in Eq. (45) vanish, leaving

$$\frac{\partial \mathcal{W}}{\partial \kappa_y} = U_2(c_1, c_2) \bar{K}^\alpha \frac{\partial \Gamma}{\partial \kappa_y} \quad (46)$$

Because $U_2 > 0$ and $\bar{K}^\alpha > 0$, the sign of the welfare effect is the sign of the effect on total factor productivity. The intuition is that the subsidy $\tau = 1 - \gamma$ sets aggregate capital at its optimal level, so that the marginal unit of capital is worth exactly its resource cost and only the allocation of capital across firms matters at the margin.

It remains to show that $\frac{\partial \Gamma}{\partial \kappa_y}$ is unaffected by the change in preferences. Using $\Gamma = \mathcal{Y}/\bar{K}^\alpha = \bar{K}^{1-\alpha}/\chi$ together with Eq. (18), the constant χ cancels and

$$\Gamma = \exp\left(\mu_z + \frac{1}{2(1-\alpha)\kappa_z}\right) \exp\left(-\frac{\alpha[(1-\alpha)(1-\rho)\kappa_x + (1-\gamma)^2(\kappa_y + \kappa_z)]}{2(1-\alpha)D^2}\right) \quad (47)$$

Total factor productivity therefore depends only on the precisions and on $(\alpha, \gamma, \rho, \mu_z, \kappa_z)$: it is independent of the cost of capital, of fiscal policy, and hence of the household's preferences. The first factor is total factor productivity under full information, Eq. (40), and the second is the loss from dispersed information.

Comparing Eq. (47) with Eq. (18), $\ln \Gamma$ and $\ln \bar{K}$ differ only by a positive multiple of the same precision-dependent term, so $\frac{\partial \Gamma}{\partial \kappa_y}$ and $\frac{\partial \bar{K}}{\partial \kappa_y}$ always share a sign. By Proposition 5 and Eq. (46),

$$\text{sgn}\left(\frac{\partial \mathcal{W}}{\partial \kappa_y}\right) = \text{sgn}\left(\frac{\kappa_y + \kappa_z}{\kappa_x} - \Delta_{\kappa_y}\right)$$

and the same argument applied to κ_x reproduces Proposition 6. The threshold characterizations of Section 4.2 and Section 4.3 therefore hold for any increasing and concave U , provided the subsidy is set at $\tau = 1 - \gamma$.

E Endogenous Disclosure

Throughout the paper the precision of the public signal is a parameter. This appendix lets firms choose it. At a date $t = 0$, before productivities and signals are realized, each firm selects the precision κ_y of the public signal about itself at a cost $\Psi(\kappa_y)$, paid in period 2 and increasing and convex in κ_y . The game then unfolds exactly as in Section 2. The household is risk neutral, so the cost of capital is fixed at $R = 1/\beta - (1 - \delta)$ by Eq. (8) and does not move with disclosure.

Write $\mathcal{Y}(\kappa_y)$ for the expected output of a firm that chooses precision κ_y . Because each firm has measure zero, a single firm's choice affects neither aggregate quantities nor R , and because

productivities are independent across a continuum of firms, when all firms choose the same κ_y the function $\mathcal{Y}(\cdot)$ is also aggregate output.

The firm's choice. By Corollary 1 the founder retains a fraction $1 - \alpha$ of output, so firm j solves

$$\max_{\kappa_y} \beta \mathbb{E}[(1 - \alpha)z_j K_j^\alpha] - \Psi(\kappa_y) \quad \Longleftrightarrow \quad \max_{\kappa_y} A^F \mathcal{Y}(\kappa_y) - \Psi(\kappa_y)$$

where

$$A^F \equiv \beta(1 - \alpha) \quad (48)$$

The socially optimal choice. Consider instead a planner who chooses κ_y to maximize welfare, taking the decentralized equilibrium that follows as given. Using Eq. (16), the planner solves

$$\max_{\kappa_y} e - [1 - \beta(1 - \delta)]\bar{K} + \beta \mathcal{Y}(\kappa_y) - \Psi(\kappa_y)$$

By Lemma 2, $\bar{K} = \mathcal{Y} \gamma \lambda / [(1 - \tau)R]$. Substituting $\lambda = \alpha$ from Corollary 1 and $R = [1 - \beta(1 - \delta)]/\beta$,

$$\bar{K} = \frac{\beta \alpha \gamma}{[1 - \beta(1 - \delta)](1 - \tau)} \mathcal{Y}$$

so that $[1 - \beta(1 - \delta)]\bar{K} = \beta \alpha \gamma \mathcal{Y} / (1 - \tau)$ and the planner's objective becomes $e + A^P \mathcal{Y}(\kappa_y) - \Psi(\kappa_y)$ with

$$A^P \equiv \beta \left(1 - \frac{\alpha \gamma}{1 - \tau} \right) \quad (49)$$

Since e is a constant, the two problems differ only in the weight placed on expected output.

Comparing the two weights. From Eq. (48) and Eq. (49),

$$A^P - A^F = \beta \alpha \left(1 - \frac{\gamma}{1 - \tau} \right)$$

so $A^P > A^F$ if and only if $1 - \tau > \gamma$, with equality exactly at the efficient subsidy $\tau = 1 - \gamma$ of Proposition 3. The gap has a direct reading. The planner values the whole of output net of the resource cost of the capital used to produce it, $\beta(\mathcal{Y} - R\bar{K})$, whereas the founder values only its own share $\beta(1 - \alpha)\mathcal{Y}$. The difference, $\beta(\alpha\mathcal{Y} - R\bar{K})$, is the surplus that accrues on the investor side net of the resource cost of their capital. The founder ignores it. At $\tau = 1 - \gamma$ that surplus is exactly zero, and the founder's private return to disclosure coincides with the social return.

Comparative statics. Both problems have the form $\max_{\kappa_y} A\mathcal{Y}(\kappa_y) - \Psi(\kappa_y)$ for a weight $A > 0$. At an interior optimum,

$$A\mathcal{Y}'(\kappa_y) = \Psi'(\kappa_y), \quad A\mathcal{Y}''(\kappa_y) - \Psi''(\kappa_y) < 0$$

the second being the second-order condition. Differentiating implicitly the first-order condition with respect to A and rearranging,

$$\frac{d\kappa_y}{dA} = \frac{\mathcal{Y}'(\kappa_y)}{\Psi''(\kappa_y) - A\mathcal{Y}''(\kappa_y)} \quad (50)$$

The denominator is positive by the second-order condition, so the sign of $d\kappa_y/dA$ is the sign of $\mathcal{Y}'(\kappa_y)$. Moreover, since Ψ is strictly increasing and $A > 0$, the first-order condition forces $\mathcal{Y}'(\kappa_y) > 0$ at any interior optimum. Disclosure is therefore increasing in the weight A .

Proposition 8. *Assume both problems have interior solutions and let $A^P > 0$. Denote by κ_y^F the firm's choice and by κ_y^P the socially optimal one. Then*

$$\kappa_y^F \leq \kappa_y^P \quad \text{according as} \quad 1 - \tau \geq \gamma$$

In particular, firms under-disclose when $\tau < 1 - \gamma$, over-disclose when $\tau > 1 - \gamma$, and disclose efficiently when $\tau = 1 - \gamma$.

Proof. Immediate from Eq. (50) and the comparison of A^F and A^P above. □